Skill Navora

TRACK 3

Power BI

ETL with Power Query, star-schema modelling, DAX measures and time intelligence, interactive reports, case studies, publishing, security and Copilot.

8 lessons | Version 2026.10 | Updated October 2026

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Contents

Every entry is a clickable link. The PDF also has bookmarks for each lesson and section.

01 Getting Started with Power BI & Your First HR Report

02 Power Query, Star Schema & Data Modelling (Adventure Cycles)

03 DAX Fundamentals: Measures, Filter Context & CALCULATE

04 DAX Time Intelligence, Ranking, Parameters & Dynamic Measures

05 Visuals, Filters, Drill-through, Tooltips, Buttons & Report Design

06 Power BI Service, Connectivity Modes, Security & Capstone

07 Microsoft Fabric: Lakehouse, Warehouse, Dataflows and Semantic Models

08 Databricks: Notebooks, Delta Lake, Spark Jobs and Unity Catalog

POWER BI · LESSON 1

Getting Started with Power BI & Your First HR Report

Understand what Power BI is and how its parts fit together, install Power BI Desktop, tour the three views, follow the ETL and analytics stages, and build a complete first report on HR data: cards, bar and column charts, a donut, slicers, a year column with DAX, groups and small multiples.

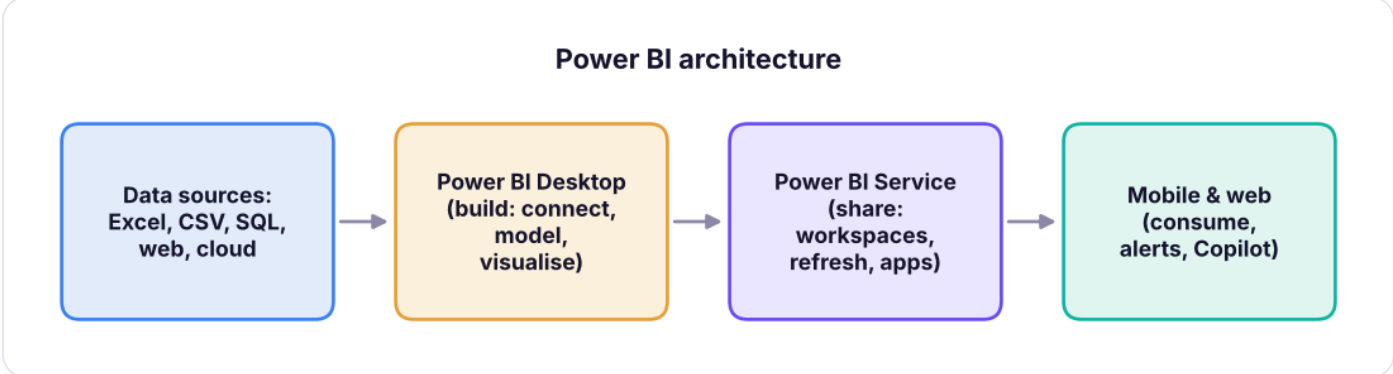
Power BI class 1 | Power BI Desktop 2026 (Copilot, Fabric, PBIP) | Beginner | 120 min

By the end of this lesson you can

- Describe Power BI Desktop, the Service and Mobile, and how data flows between them
- Explain ETL and the stages of the data-analytics process, and map them to Power BI tools
- Navigate Report, Table and Model views, and the Fields, Visualizations and Filters panes
- Load a CSV, check data types and build cards, bar/column charts, a matrix, a donut and slicers
- Add a calculated column with DAX (YEAR), create data groups, and use small multiples

What is Power BI?

Power BI is Microsoft's business-intelligence platform: it connects to data, cleans and models it, and turns it into interactive reports and dashboards that people can share. In 2026 it is part of **Microsoft Fabric**, Microsoft's all-in-one analytics platform, and it is one of the most requested tools in analyst job posts, alongside Excel, SQL and Tableau.



The Power BI components. You build in Desktop, publish to the Service, and consume on the web or mobile.

Component	What it is	Who uses it
Power BI Desktop	Free Windows application to build data models and reports (`.pbix`)	Analysts, developers
Power BI Service (app.powerbi.com)	Cloud site to publish, share, schedule refresh, create dashboards and apps	Analysts and report consumers
Power BI Mobile	iOS/Android apps to view reports and get alerts	Managers on the go
Power BI Report Builder	Paginated, print-ready reports (invoices, statements)	Report developers
On-premises data gateway	Secure bridge so the cloud can refresh data from your own servers	Admins
Power BI Report Server	Self-hosted option for organisations that cannot use the cloud	IT
Copilot	AI assistant that writes DAX, builds report pages and summarises insights (needs a Fabric/Premium capacity)	Everyone

> **TIP** **Report vs dashboard**
>
> In Power BI a **report** is one or more pages of interactive visuals built on a data model. A **dashboard** (Service only) is a single page of **pinned tiles** from different reports. Many teams use reports only, or share them as an **app**.

Licences in brief

Licence	Use
Power BI Desktop	Free: build and save locally
Free (Service)	Personal use in "My workspace"; cannot share with others
Pro	Publish and share with other Pro users (about US$14 per user per month)
Premium Per User / Fabric capacity	Larger models, AI features, share with free users

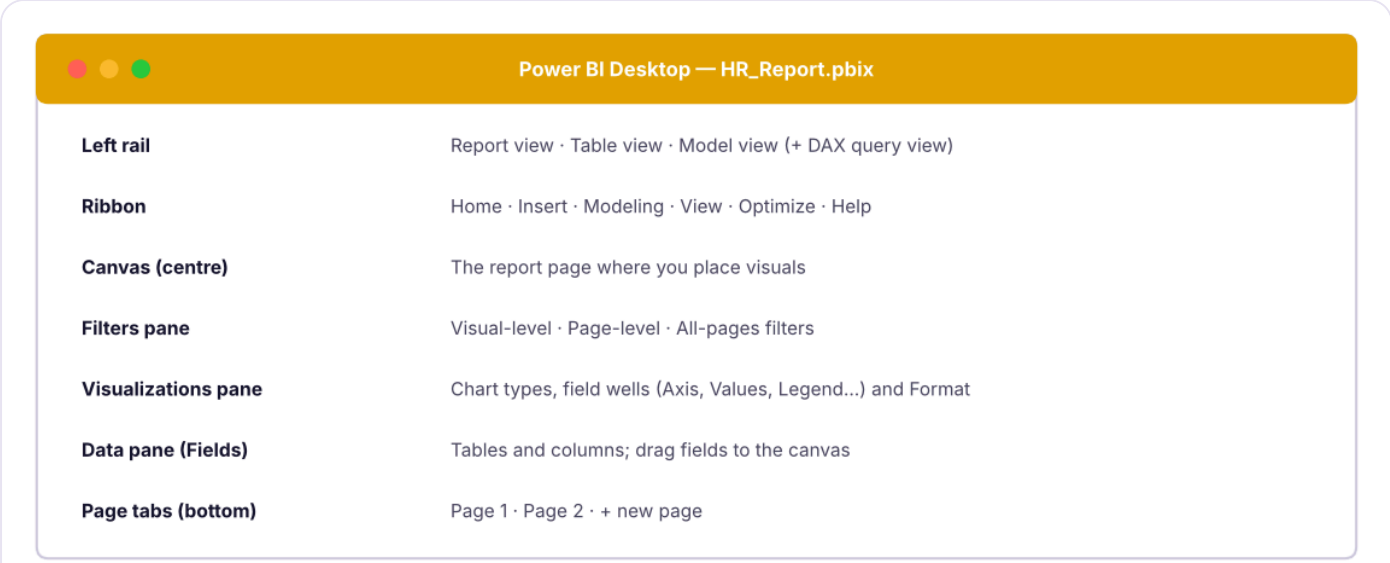
(Prices change; check Microsoft's current pricing page.)

Installing Power BI Desktop

STEP BY STEP **Install**

1. **Easiest:** open the **Microsoft Store**, search "Power BI Desktop" and click Install (it updates itself every month).
2. Or download it from the Microsoft Power BI website → Download → Power BI Desktop.
3. Requirements: Windows 10/11 (64-bit), 4 GB+ RAM (8 GB+ recommended). **There is no Mac version**: Mac users can use the Service in a browser, a Windows virtual machine, or Windows 365.
4. Start the app and sign in with a work or school account (optional for building locally).

The three views

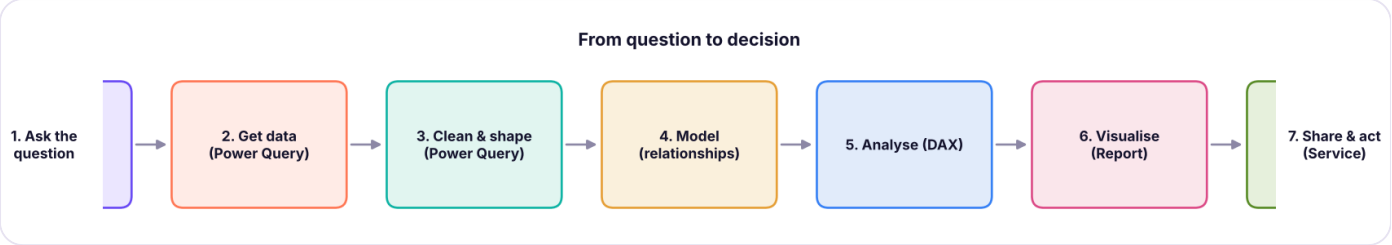


The Power BI Desktop window: left rail with three views, canvas in the middle, panes on the right.

View	Icon	Use it to
Report	bar chart	Design the report pages
Table (Data)	grid	Inspect the data and create calculated columns and measures
Model	diagram	See and edit **relationships** between tables
DAX query view	`{ }`	Run DAX queries and test measures

ETL and the analytics process

ETL = **E**xtract (get data from sources), **T**ransform (clean and reshape it), **L**oad (store it in the model). In Power BI, Extract and Transform are done in **Power Query**, and Load puts the result into the data model. (You met Power Query in Excel Lesson 12: it is the same engine.)



The data analytics stages, mapped to Power BI.

Always start with the **question**: "Why are people leaving?" decides which data and visuals you need. The worst Power BI reports are tours of every chart type with no question.

Your first report: HR attrition

CASE STUDY **The brief**

HR wants a one-page report on the company's 1,500 employees: how many there are, how many left (attrition), where attrition is highest, and whether overtime, job satisfaction or income play a role. Use `hr_employees.csv`.

The data (first rows):

Python

```python
import pandas as pd
hr = pd.read_csv("hr_employees.csv", parse_dates=["hire_date"])
print(hr.shape)
hr.head(5)
```

OUTPUT

```
(1500, 15)
```

	emp_id	gender	age	department	job_title	education	hire_date	years_at_c
0	E00001	Male	39	Operations	Operations Manager	Bachelor's	2025-05-26	0
1	E00002	Female	45	IT	Tester	Bachelor's	2020-10-17	5
2	E00003	Male	41	IT	Tester	Bachelor's	2019-05-06	6
3	E00004	Male	21	R&D	Data Scientist	Bachelor's	2025-08-23	0
4	E00005	Male	41	Sales	Sales Executive	Master's	2015-08-24	10

5 rows × 15 columns

Step 1: get the data

STEP BY STEP **Load the CSV**

1. Home → **Get data → Text/CSV** → choose `hr_employees.csv`.
2. The preview shows the columns and detected data types. Click **Transform Data** (not Load) to check types in Power Query.
3. Confirm: `age`, `years_at_company`, `monthly_income`, `job_satisfaction`, `performance_rating` are whole numbers; `hire_date` is **Date**; the rest are text.
4. Home → **Close & Apply**.

In the Data pane you now see the table `hr_employees` with a **Σ** next to numeric fields (they can be summed) and no symbol next to text fields.

Step 2: KPI cards

Drag fields onto the canvas, then choose the **Card** visual:

- **Employees** = Count of `emp_id`
- **Attrition (leavers)**: use a measure (Lesson 3 explains measures): `Leavers = CALCULATE ( COUNTROWS ( hr_employees ), hr_employees[attrition] = "Yes" )`
- **Attrition %** = `DIVIDE ( [Leavers], COUNTROWS ( hr_employees ) )`, formatted as a percentage
- **Average monthly income** = Average of `monthly_income`

Python

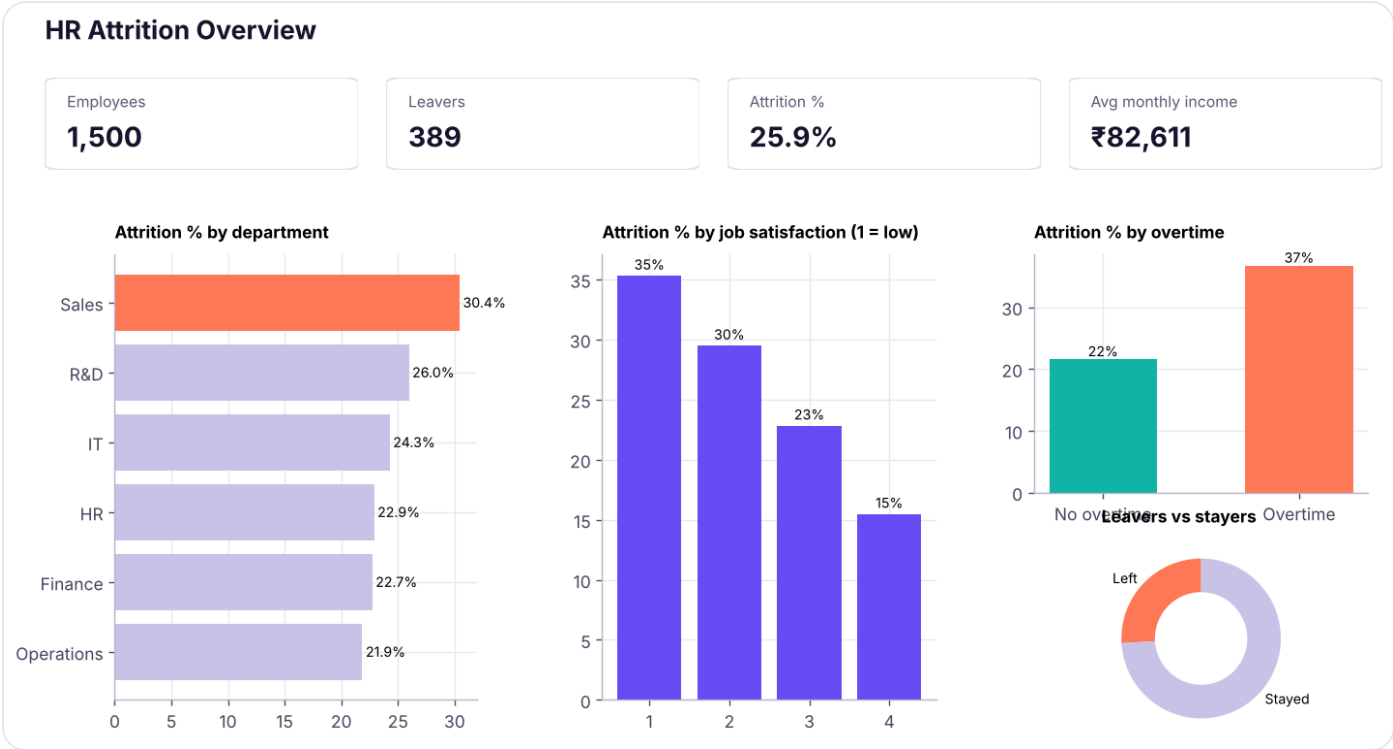
```python
n = len(hr); leavers = (hr.attrition == "Yes").sum()
print(f"Employees      : {n:,}")
print(f"Leavers        : {leavers:,}")
print(f"Attrition %    : {leavers / n:.1%}")
print(f"Avg income (₹) : {hr.monthly_income.mean():,.0f}")
```

OUTPUT

```
Employees      : 1,500
Leavers        : 389
Attrition %    : 25.9%
Avg income (₹) : 82,611
```

Step 3: bar, column and donut charts

- **Clustered bar chart**: Y-axis `department`, X-axis = `Attrition %` → ranks departments.
- **Clustered column chart**: X-axis `job_satisfaction`, Y-axis `Attrition %`.
- **Donut chart**: Legend `attrition`, Values = count of `emp_id`.
- **Matrix**: Rows `department`, Columns `overtime`, Values `Attrition %` (turns into a heat map with conditional formatting, Lesson 5).
- **Slicers**: `gender`, `education`, `city`.



The finished HR page, drawn from the real data: KPI cards, attrition by department, by satisfaction and overtime, and the leaver/stayer donut.

Step 4: a calculated column with DAX

We want to see joining trends by year, but the table has only `hire_date`. Add a **calculated column**: Table view (or Modeling) → **New column**:

DAX

```
Hire Year = YEAR ( hr_employees[hire_date] )
```

A calculated column is evaluated **row by row when the data loads** and stored in the model, just like a new column in Excel. Now a line/column chart of **headcount by hire year** works:

Python

```
hr["hire_year"] = hr.hire_date.dt.year
by_year = hr.groupby("hire_year").agg(hired=("emp_id", "count"), left=("left", "sum"))
by_year["attrition_%"] = (by_year.left / by_year.hired * 100).round(1)
by_year.tail(8)
```

OUTPUT

	hired	left	attrition_%
hire_year			
2019	82	16	19.5
2020	85	20	23.5
2021	120	22	18.3
2022	136	35	25.7
2023	196	60	30.6
2024	234	71	30.3
2025	284	93	32.7
2026	64	17	26.6

8 rows × 3 columns

Step 5: data groups and bins

Groups combine values into custom categories, such as the "age band" below, without changing the source data:

- **Bins** (numbers): right-click the field → **New group** → Group type *Bin* → bin size 5 → creates `age (bins)`.
- **List groups** (text): select several values → right-click → **Group** → rename, for example group *Mumbai* and *Pune* as "Maharashtra".

For flexible logic, a calculated column is more transparent:

DAX

```
Age Band =
SWITCH (
    TRUE (),
    hr_employees[age] < 25, "Under 25",
    hr_employees[age] < 35, "25-34",
    hr_employees[age] < 45, "35-44",
    "45 and over"
)
```

Python

```
bands = pd.cut(hr.age, [0, 24, 34, 44, 120], labels=["Under 25", "25-34", "35-44", "45 and over"])
hr.groupby(bands, observed=False).agg(employees=("emp_id", "count"), attrition_pct=("left", lambda s: round(s.mean() * 100, 1)))
```

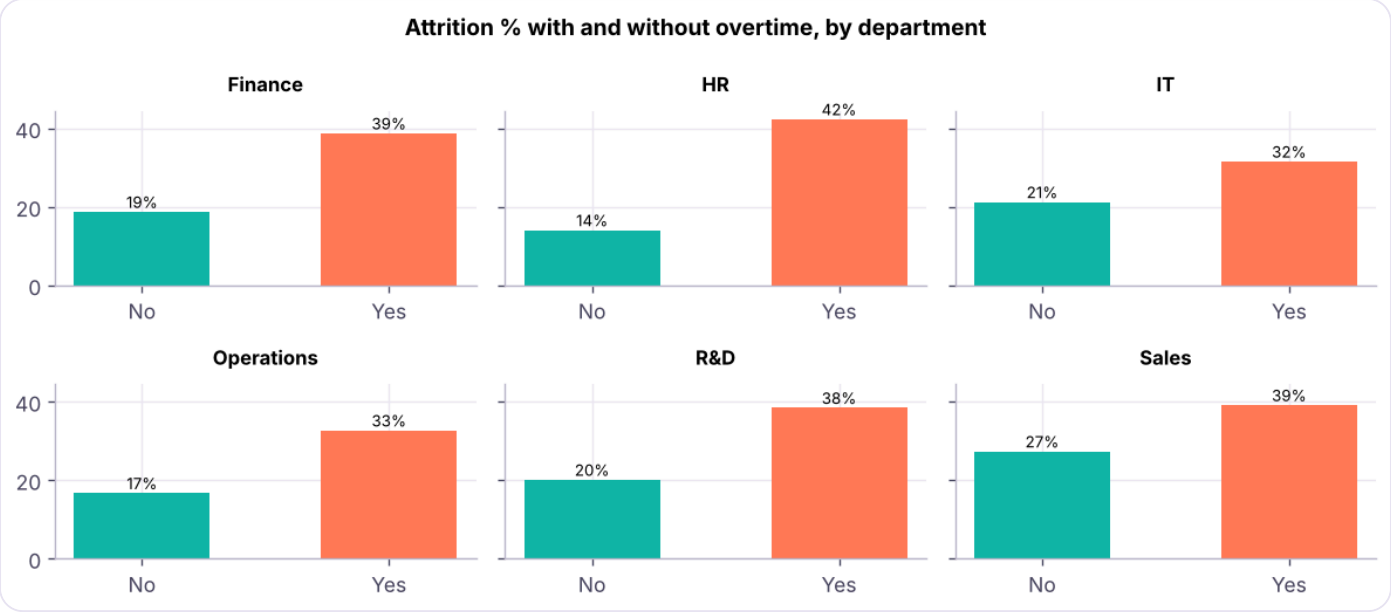
OUTPUT

	employees	attrition_pct
age		
Under 25	148	35.1
25–34	624	26.9
35–44	574	23.5
45 and over	154	22.1

4 rows × 2 columns

Step 6: small multiples

Small multiples split one visual into a grid of identical mini-charts by a field. Drag `department` into the visual's **Small multiples** well: you get one small column chart of attrition by satisfaction for every department. This lets you compare patterns across groups at a glance without a cluttered legend.



Small multiples: attrition % by overtime for each department. The same pattern repeats everywhere, which says overtime is a company-wide issue, not a one-department issue.

What does the report tell us?

Python

```
t = hr.groupby("overtime").left.agg(["count", "mean"]); t["mean"] = (t["mean"] *
100).round(1)
t.columns = ["employees", "attrition_%"]
print(t.to_string()); print()
low = hr[hr.job_satisfaction ⩽ 2].left.mean() * 100; high = hr[hr.job_satisfaction ⩾
3].left.mean() * 100
print(f"Attrition for satisfaction 1-2: {low:.1f}%   vs   3-4: {high:.1f}%")
```

OUTPUT

```
           employees  attrition_%
overtime
No              1067         21.6
Yes              433         36.7

Attrition for satisfaction 1-2: 32.5%   vs   3-4: 19.4%
```

Insight sentences for a manager: *overtime workers leave far more often than others; low satisfaction scores go with higher attrition; Sales has the highest attrition among departments*. Notice the careful wording: these are **associations** in the data, not proof of cause. (The ML track shows how to test them properly.)

Good habits from day one

1. **Name things clearly**: `Attrition %`, not `Measure 3`; rename pages ("HR Overview").

2. **Put measures in a dedicated table** later (Lesson 3) and format them (percentage, thousands separator).
3. **Check data types** in Power Query before loading.
4. **One idea per page**; the title should say the insight.
5. **Save often** (`.pbix`), and use clear version names.
6. **Do not use pie charts** with many slices; prefer bars (ranked) and use donuts only for 2–3 parts.

2026 UPDATE

Power BI in 2026

- **Copilot** can create a report page from a prompt ("Build an attrition overview for HR"), write DAX measures and summarise a page. It needs a paid Fabric/Premium capacity and your data to be well-modelled and named.
- **Power BI Project (PBIP)** format stores the report as text files, so teams can use **Git** and code review.
- **TMDL** (Tabular Model Definition Language) and **DAX query view** make models easier to script and test.
- **Direct Lake** (Fabric) reads Delta tables directly with import-like speed. (See Lesson 6.)

Summary

KEY TAKEAWAYS

- Power BI = Desktop (build) + Service (share) + Mobile (consume), now inside Microsoft Fabric.
- ETL: **Extract and Transform in Power Query, Load into the model**; analytics stages run from question to decision.
- Three views: **Report**, **Table**, **Model**; build with the Fields, Visualizations and Filters panes.
- A **card** shows a KPI; bar/column charts compare categories; a donut shows 2–3 parts; a matrix is a pivot table; slicers filter.
- A DAX **calculated column** (`YEAR`, `SWITCH`) is computed row by row at load time; **groups/bins** categorise without changing the data; **small multiples** repeat a chart per category.

Quick quiz

6 questions · tap an answer to check it

1. In Power BI, where do you perform the Extract and Transform steps of ETL?

- **A** Report view
- **B** Power Query
- **C** The Service
- **D** The Model view

Answer: B. Power Query connects to sources and transforms data; Close & Apply loads it into the model.

2. Which view shows relationships between tables?

- **A** Report view
- **B** Table view
- **C** Model view
- **D** Filters pane

Answer: C. Relationships are created and edited in Model view.

3. Power BI Desktop is available for:

- **A** Windows and Mac
- **B** Windows only
- **C** Mac only
- **D** Browsers only

Answer: B. Desktop is Windows-only; Mac users use the Service or a Windows virtual machine.

4. What does a calculated column do?

A Runs only when a slicer changes

B Computes a value for every row when the data is loaded and stores it in the model

C Replaces a measure

D Changes the source file

Answer: B. Calculated columns are row-by-row and stored; measures calculate on the fly.

5. Which visual repeats the same chart for every value of a field?

A Matrix

B Slicer

C Small multiples

D Card

Answer: C. Small multiples create a grid of mini-charts split by a category.

6. Which licence do you need to share a report with colleagues who also have Pro licences?

A None, Desktop is enough

B Free Service licence

C Power BI Pro (or Premium Per User)

D Gateway licence

Answer: C. Sharing in the Service requires Pro for creators and viewers (unless the workspace is on Premium/Fabric capacity).

EXERCISE

Build the HR page

Build the page shown above in Power BI Desktop with `hr_employees.csv`. Then add: (1) a slicer for `city` and `gender`; (2) a matrix of attrition % by `job_title` and `education`; (3) an `Age Band` column and a column chart of attrition % by band; (4) a title text box that states your main finding.

Show solution (worked answer)

(1) Slicer visual, fields `city`, `gender` (Format → Slicer settings → Style: Tile or Dropdown). (2) Matrix: Rows `job_title`, Columns `education`, Values `Attrition %` (turn on Cell elements → Background colour for a heat map). (3) Use the `SWITCH` formula in this lesson; chart X-axis `Age Band`, Y-axis `Attrition %`; sort the axis in the right order using a sort-by-column. (4) Insert → Text box, for example "Overtime is linked to about 15 more attrition points company-wide." Verify the claim against your visuals before you publish.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between Power BI Desktop and Service? Desktop is the free authoring tool for models and reports on Windows; the Service is the cloud platform to publish, share, refresh on a schedule, build dashboards and apps, and apply security.

Q: What is ETL and where is it done in Power BI? Extract, Transform, Load: Power Query connects to sources and applies transformation steps; the result is loaded into the VertiPaq in-memory model.

Q: Calculated column vs measure? A calculated column is evaluated per row at refresh time and stored (uses memory); a measure is evaluated at query time in the current filter context (uses CPU). Use measures for aggregations such as sums and ratios.

Q: Report vs dashboard? A report is multi-page and interactive, built on one model. A dashboard is a single page of pinned tiles from several reports, available only in the Service.

POWER BI · LESSON 2

Power Query, Star Schema & Data Modelling (Adventure Cycles)

Prepare and model data the professional way. Clean data in Power Query, load a star schema (fact and dimension tables), create one-to-many relationships, understand filter direction, build and mark a date table, create hierarchies, and avoid the most common modelling mistakes, using the Adventure Cycles case study.

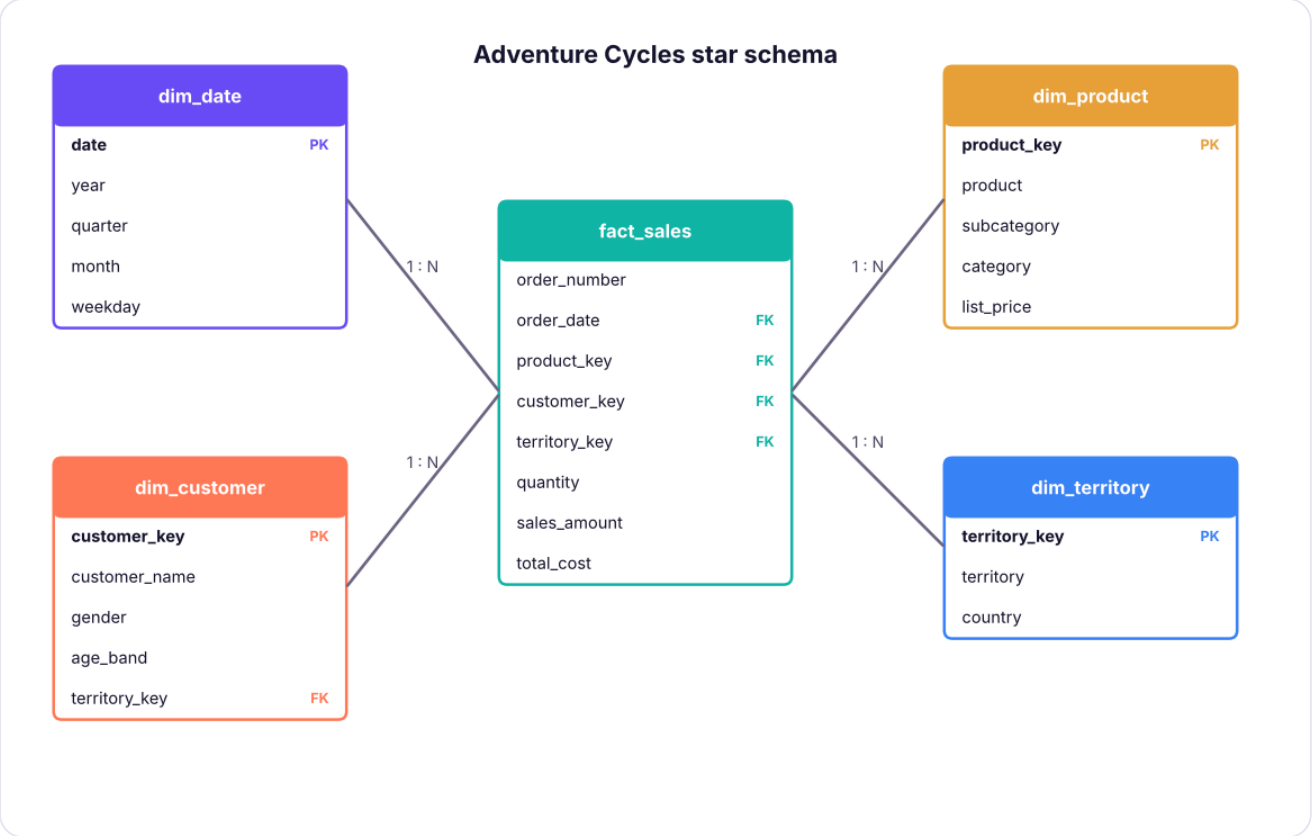
Power BI class 3 | Power BI Desktop 2026 | Intermediate | 140 min

By the end of this lesson you can

- Apply common Power Query steps and read the Applied Steps and M code
- Explain fact tables, dimension tables and the star schema, and why they beat one wide table
- Create relationships with the right cardinality and filter direction, and explain how filters flow
- Build a date table, mark it as a date table and add hierarchies
- Diagnose common modelling problems (many-to-many, ambiguous paths, missing dates)

Why modelling matters more than visuals

Beautiful visuals on a bad model give wrong or slow answers. In Power BI the **data model** decides what numbers mean: how tables relate, which table filters which, and how fast DAX runs. The golden rule: **model as a star schema**.



A star schema: one central fact table of events/numbers surrounded by descriptive dimension tables. Filters flow from dimensions (one side) to the fact (many side).

	Fact table	Dimension table
Describes	Events or transactions (a sale, a click, a shipment)	The "who, what, where, when" of those events
Rows	Many (millions)	Fewer (thousands)
Columns	Keys + numeric measures (quantity, amount, cost)	Keys + descriptive attributes (name, category, city)
Changes	Grows constantly	Changes rarely
Example	`fact_sales`	`dim_product`, `dim_customer`, `dim_date`

Why a star beats one big flat table: smaller file (text stored once), faster visuals, one place to fix a name, correct filtering across several facts, and simpler DAX. A **snowflake** (dimensions normalised into sub-dimensions) works but adds joins; prefer a star in Power BI.

The Adventure Cycles dataset

A fictional bicycle company sells bikes, components, clothing and accessories in India, the UAE and Singapore. The model has one fact table and four dimensions (plus the date table):

Python

```
import pandas as pd
D = "adventure_cycles/"
fact = pd.read_csv(D + "fact_sales.csv", parse_dates=["order_date"]); prod = pd.read_csv(D
+ "dim_product.csv")
cust = pd.read_csv(D + "dim_customer.csv"); terr = pd.read_csv(D + "dim_territory.csv");
dates = pd.read_csv(D + "dim_date.csv", parse_dates=["date"])
for name, df in [("fact_sales", fact), ("dim_product", prod), ("dim_customer", cust),
("dim_territory", terr), ("dim_date", dates)]:
    print(f"{name:14s} {df.shape[0]:>6,} rows × {df.shape[1]} columns")
fact.head(4)
```

OUTPUT

```
fact_sales      9,000 rows × 9 columns
dim_product        45 rows × 6 columns
dim_customer    1,200 rows × 5 columns
dim_territory       6 rows × 3 columns
dim_date        1,096 rows × 6 columns
```

	order_number	order_date	product_key	customer_key	territory_key	quantity	ur
0	SO50387	2023-01-01	11	103	1	3	1:
1	SO55797	2023-01-01	32	798	1	3	2(
2	SO51236	2023-01-01	9	750	5	1	1:
3	SO55082	2023-01-01	13	759	1	1	7!

4 rows × 9 columns

Power Query: prepare each table

Open **Transform Data**. Typical steps for each table (all recorded in **Applied Steps**):

Table	Steps
`fact_sales`	Set types (`order_date` Date; `quantity` Whole number; amounts **Fixed decimal number**), remove blank rows, add `Profit = [sales_amount] - [total_cost]` as a custom column, keep only needed columns
`dim_product`	Trim/clean text, set `list_price` and `standard_cost` as decimals, remove duplicates on `product_key`
`dim_customer`	Replace `M`/`F` with Male/Female, remove duplicates on `customer_key`
`dim_territory`	Check there is one row per `territory_key`
`dim_date`	Set `date` to Date; sort `month` by `month_no` later in the model

Query settings (right pane) shows each step; click any step to see the data at that point, and delete a step to undo it. Rename queries (`Sales`, `Product`…) because those names become table names in the model.

Common transformations (from class and beyond)

Goal	Where	Notes
Promote headers, set types	Home	Always check types: wrong types break relationships
Remove duplicates / blank rows	Home → Remove rows	Needed for dimension keys
Split column by delimiter	Transform → Split Column	e.g. `IN-2025-00042`
Replace values, trim, clean, format	Transform	
Unpivot columns	Transform → Unpivot	Turn Jan, Feb, Mar columns into rows (Excel Lesson 12)
Merge queries	Home → Merge	A join; use to bring a category into the fact (but prefer relationships)
Append queries	Home → Append	Stack monthly files
Group by	Transform → Group By	Pre-aggregate large tables
Conditional / Custom column	Add Column	M formulas
Reference vs Duplicate	Right-click query	*Reference* creates a new query that starts from the output of another (changes flow through); *Duplicate* copies the steps
Parameters	Home → Manage Parameters	A variable such as a file path or a start date, used in queries (Lesson 4)
Disable load	Right-click query	Keep helper queries out of the model

What is "query folding"?

When the source is a database (SQL Server, Snowflake), Power Query **translates your steps into SQL** and lets the database do the work, instead of downloading everything first. A step that cannot be translated "breaks folding" and everything after it runs locally. Check by right-clicking a step → *View Native Query*. Keep foldable steps (filter, select columns, merge, group) **before** non-foldable ones.

Power Query M

```
let
    Source   = Csv.Document(File.Contents("C:\Data\fact_sales.csv"), [Delimiter=",",
Encoding=65001]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Typed    = Table.TransformColumnTypes(Promoted, {{"order_date", type date},
{"quantity", Int64.Type},
                  {"sales_amount", Currency.Type}, {"total_cost", Currency.Type}}),
    Profit   = Table.AddColumn(Typed, "profit", each [sales_amount] - [total_cost],
Currency.Type)
in
    Profit
```

Relationships

A **relationship** links a column in one table to a column in another so that filters can flow between them. In **Model view**, drag `dim_product[product_key]` onto `fact_sales[product_key]`. Power BI usually detects relationships on same-named columns (Home → Manage relationships → Autodetect).

Cardinality

Type	Meaning	Use
One-to-many (1:*)	A key is unique in one table, repeated in the other	The standard: dimension → fact
One-to-one (1:1)	Unique on both sides	Splitting a wide table
Many-to-many (*:*)	Duplicates on both sides	Avoid; use a bridge table

Check uniqueness before creating a one-to-many: if the "one" side has duplicates Power BI will complain.

Python

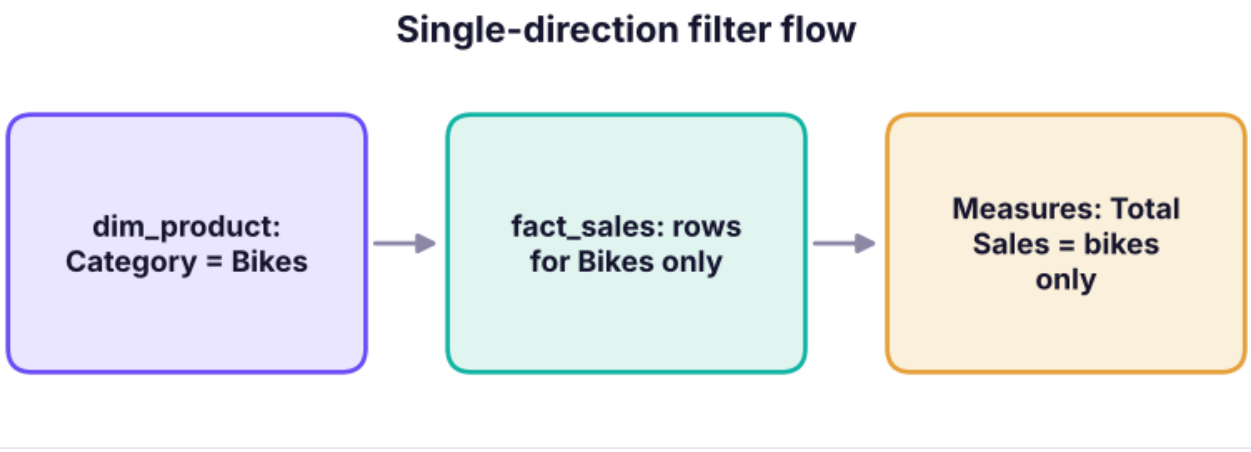
```
checks = {"dim_product[product_key]": prod.product_key, "dim_customer[customer_key]":
cust.customer_key, "dim_territory[territory_key]": terr.territory_key, "dim_date[date]":
dates.date}
for name, col in checks.items(): print(f"{name:30s} unique: {col.is_unique}")
orphans = {"product_key": ~fact.product_key.isin(prod.product_key), "customer_key":
~fact.customer_key.isin(cust.customer_key), "territory_key":
~fact.territory_key.isin(terr.territory_key)}
for k, m in orphans.items(): print(f"fact rows with no matching {k}: {m.sum()}")
```

OUTPUT

```
dim_product[product_key]       unique: True
dim_customer[customer_key]     unique: True
dim_territory[territory_key]   unique: True
dim_date[date]                 unique: True
fact rows with no matching product_key: 0
fact rows with no matching customer_key: 0
fact rows with no matching territory_key: 0
```

Unique keys on the "one" side and no orphans on the "many" side: the model is safe.

Filter direction



Filters flow along relationships from the one side to the many side (single direction). Selecting Bikes in dim_product filters fact_sales, which then filters the other dimensions only if bi-directional filtering is on.

Setting	Behaviour	Advice
Single (dimension → fact)	A selection in a dimension filters the fact	Default and best
Both (bi-directional)	Filters also flow back from fact to dimension	Use only when needed (many-to-many, "show only customers who bought"); can cause ambiguity and slowness
Active vs inactive relationship	Only one active path between two tables; an inactive one can be switched on in DAX with `USERELATIONSHIP`	e.g. Order Date and Ship Date both link to the date table

Seeing the model work

Total sales by product category and by country, using the relationships (a join in pandas):

Python

```python
m = fact.merge(prod, on="product_key").merge(terr, on="territory_key")
m["profit"] = m.sales_amount - m.total_cost
out = m.pivot_table(index="category", columns="country", values="sales_amount",
aggfunc="sum", margins=True, margins_name="Total") / 1e6
out.round(1).rename_axis("Sales (₹ million)")
```

OUTPUT

country	India	Singapore	UAE	Total
Sales (₹ million)				
Accessories	6.7	1.7	1.3	9.8
Bikes	320.6	74.8	71.3	466.8
Clothing	6.0	1.3	1.1	8.5
Components	73.5	18.8	17.2	109.6
Total	406.9	96.7	91.0	594.6

5 rows × 4 columns

In Power BI you get this with a matrix: Rows `category`, Columns `country`, Values `SUM(sales_amount)`. No join code needed: the relationships do it.

The date table

Time intelligence (year-to-date, same period last year, Lesson 4) requires a **proper date table**: one row per day with **no gaps**, covering all dates in the data, and **marked as a date table**.

DAX

```
Date =
VAR MinDate = DATE ( YEAR ( MIN ( fact_sales[order_date] ) ), 1, 1 )
VAR MaxDate = DATE ( YEAR ( MAX ( fact_sales[order_date] ) ), 12, 31 )
RETURN
ADDCOLUMNS (
    CALENDAR ( MinDate, MaxDate ),
    "Year",       YEAR ( [Date] ),
    "Quarter",    "Q" & QUARTER ( [Date] ),
    "Month No",   MONTH ( [Date] ),
    "Month",      FORMAT ( [Date], "mmm" ),
    "Year-Month", FORMAT ( [Date], "yyyy-mm" ),
    "Weekday",    FORMAT ( [Date], "ddd" )
)
```

STEP BY STEP

Set it up

1. Model → **New table** → paste the DAX above (or load `dim_date.csv`).
2. Select the table → **Table tools → Mark as date table** → choose the `Date` column. Power BI validates that dates are unique and continuous.
3. Create the relationship `Date[Date]` → `fact_sales[order_date]` (one-to-many).
4. **Sort `Month` by `Month No`** (select `Month` → Column tools → Sort by column) so April does not come before January.
5. Hide the raw `order_date` in the fact table so report authors use the date table.

Python

```
print("Fact order dates :", fact.order_date.min().date(), "to",
fact.order_date.max().date())
print("Date table       :", dates.date.min().date(), "to", dates.date.max().date(), "|
rows:", len(dates))
full = pd.date_range(dates.date.min(), dates.date.max())
print("Continuous (no gaps):", len(full) == len(dates) and (dates.date.sort_values().values
== full.values).all())
dates.head(4)
```

OUTPUT

```
Fact order dates : 2023-01-01 to 2025-12-30
Date table       : 2023-01-01 to 2025-12-31 | rows: 1096
Continuous (no gaps): True
```

	date	year	quarter	month_no	month	weekday
0	2023-01-01	2023	Q1	1	Jan	Sun
1	2023-01-02	2023	Q1	1	Jan	Mon
2	2023-01-03	2023	Q1	1	Jan	Tue
3	2023-01-04	2023	Q1	1	Jan	Wed

4 rows × 6 columns

WATCH OUT **Do not rely on auto date/time**

Power BI can create hidden date tables automatically ("Auto date/time"). They bloat the model and cannot be customised or shared. Turn it off (File → Options → Data Load → Time intelligence) and always use your own marked date table.

Hierarchies

A **hierarchy** lets users drill down through levels with one click. In the Data pane, right-click `category` in `dim_product` → **Create hierarchy**, then drag `subcategory` and `product` onto it. Visuals with a hierarchy get drill-down buttons (arrows at the top-right of the visual): from category → sub-category → product. The date table's `Year → Quarter → Month → Date` is the classic one.

Common modelling problems

Symptom	Likely cause	Fix
Totals repeat the same number for every category	No relationship between the table you filter and the fact	Create the relationship
"A circular dependency was detected" / ambiguous paths	Two paths between two tables	Make one inactive, or remove the bi-directional filter
Many-to-many warning	Duplicate keys on both sides	Build a clean dimension with unique keys, or a bridge table
Blank row ("(Blank)") appears in visuals	Fact rows whose key is missing in the dimension (orphans)	Add the missing rows to the dimension, or fix keys
YTD / last-year measures give wrong or blank results	Date table not marked, has gaps, or not related	Fix the date table
Slow model, big file	Wide tables, high-cardinality text columns, auto date/time	Star schema, remove unused columns, reduce precision, disable auto date/time
Months sort alphabetically	Missing sort-by-column	Sort `Month` by `Month No`

TIP **Best-practice checklist**

1. Star schema with surrogate integer keys. 2. One-to-many, single-direction relationships. 3. One marked date table. 4. Hide keys and technical columns from report view. 5. Disable auto date/time. 6. Remove columns you do not use. 7. Name tables and columns for business users. 8. Put measures in a separate `_Measures` table (Lesson 3).

DO IT WITH AI **Do it with AI**

Ask Copilot or any assistant: *"Review this list of tables and columns and propose a star schema with a fact table and dimensions, the keys, and the relationships"* and paste your column lists. Compare its answer with the checklist above, and verify key uniqueness yourself in Power Query (Column profile → *Distinct* equals *Count*).

Summary

KEY TAKEAWAYS

- Model as a **star schema**: facts (events and numbers) surrounded by dimensions (descriptions), joined by **one-to-many, single-direction** relationships.
- Power Query prepares each table: types, cleaning, unpivot, merge/append, group, parameters; **query folding** pushes work to the source.
- Filters flow from the **one side** (dimension) to the **many side** (fact); use bi-directional filters sparingly; one active relationship per table pair, others inactive (`USERELATIONSHIP`).
- A continuous, **marked date table** is required for time intelligence; sort month names by month number; turn off auto date/time.
- Hierarchies enable drill-down; most "wrong totals" come from missing relationships, duplicate keys or orphans.

Quick quiz

6 questions · tap an answer to check it

1. In a star schema, which table typically contains the numeric measures such as quantity and sales amount?

A Dimension table

B Fact table

C Date table

D Parameter table

Answer: B. Facts hold events and measures; dimensions hold descriptions.

2. A one-to-many relationship between dim_product and fact_sales means:

A A product appears once in sales

B Each product_key appears once in dim_product and many times in fact_sales

C Both tables have unique keys

D The relationship is bidirectional

Answer: B. The "one" side must have unique keys.

3. Why is a bi-directional relationship risky?

A It is not allowed

B It can create ambiguous filter paths and slow performance

C It deletes data

D It only works with dates

Answer: B. Use it only when you must (for example many-to-many scenarios).

4. What does "Mark as date table" require?

A A column of text

B A unique, continuous Date column with no gaps

C A relationship to every table

D A measure

Answer: B. Time-intelligence functions need a valid contiguous date table.

5. Months show as Apr, Aug, Dec, Feb... in alphabetical order. Fix?

- **A** Rename the months
- **B** Use a pie chart
- **C** Sort the Month column by the Month No column
- **D** Delete the date table

Answer: C. Sort by column controls the display order.

6. What does Power Query's "query folding" do?

- **A** Compresses the PBIX file
- **B** Translates steps into the source system's query (e.g. SQL) so the database does the work
- **C** Hides the query
- **D** Merges two queries

Answer: B. Folding improves refresh performance for database sources.

EXERCISE

Model it yourself

Load the five Adventure Cycles CSV files, set types, create the four relationships, mark `dim_date` as the date table, hide key columns, create a Category → Subcategory → Product hierarchy, and build a matrix of sales by `category` (rows) and `year` (columns).

Show solution (worked answer)

In Power Query set `order_date` and `date` to Date. In Model view drag `product_key`, `customer_key`, `territory_key` from each dimension to the fact, and `date` to `order_date`. Table tools → Mark as date table → `date`. Right-click the key columns → Hide in report view. Create the hierarchy by right-clicking `category`. In Report view add a Matrix with Rows `dim_product[category]`, Columns `dim_date[year]`, Values `Sum of sales_amount`. Compare with the pandas pivot: Bikes should be about ₹46.7 crore of the total ₹59.5 crore.

INTERVIEW CORNER

Questions you may be asked

Q: What is a star schema and why use it in Power BI? One fact table joined to several dimension tables with one-to-many relationships. It compresses well, filters predictably, and makes DAX simple and fast.

Q: How do filters propagate in a Power BI model? From the one side to the many side along active relationships, in the filter direction. Bi-directional relationships also filter from many back to one.

Q: Why do you need a date table? Built-in time-intelligence DAX (YTD, SAMEPERIODLASTYEAR) needs a continuous, marked date table; it also gives you year/quarter/month columns, fiscal calendars and holidays.

Q: How do you handle two dates in the fact table (order date and ship date)? Role-playing dimension: one date table with one active relationship (order date) and one inactive (ship date) activated with `USERELATIONSHIP` in specific measures, or two copies of the date table.

POWER BI · LESSON 3

DAX Fundamentals: Measures, Filter Context & CALCULATE

Learn DAX, the formula language of Power BI. Understand calculated columns vs measures, row context vs filter context, write core aggregations and iterators (SUMX), master CALCULATE and FILTER, ALL/REMOVEFILTERS and ALLSELECTED, handle division safely, use variables, and build a library of business measures.

Power BI classes 2–3 | Power BI Desktop 2026 | Intermediate | 150 min

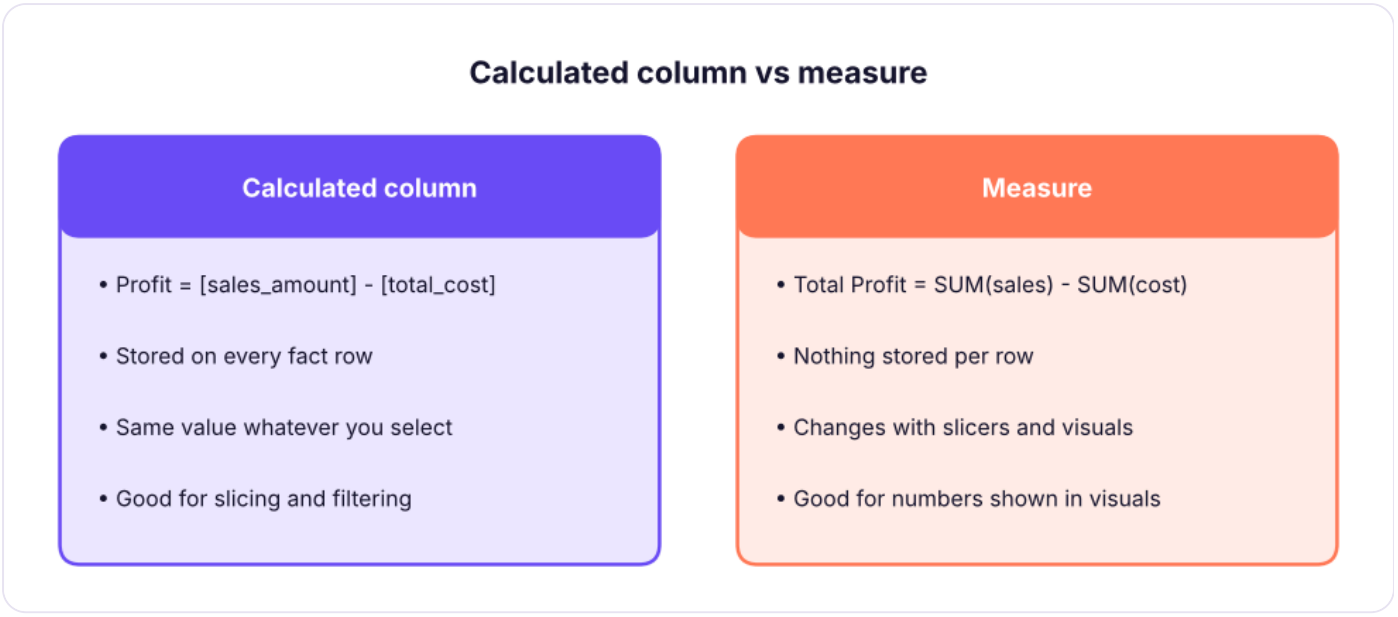
By the end of this lesson you can

- Explain the difference between a calculated column and a measure, and when to use each
- Describe row context and filter context and predict what a measure returns in a visual
- Write aggregation measures and iterators (SUM, COUNTROWS, DISTINCTCOUNT, SUMX, AVERAGEX)
- Use CALCULATE with filter arguments, ALL/REMOVEFILTERS and ALLSELECTED to build ratios and percent-of-total
- Use variables, DIVIDE, IF/SWITCH and display folders to write clean, reusable measures

What is DAX?

DAX (Data Analysis Expressions) is the formula language used in Power BI, Excel's Data Model and Analysis Services. It looks like Excel formulas, but it works on **columns and tables** and recalculates for whatever the user selects in a visual. DAX is the skill that separates a report builder from a Power BI developer.

	Calculated column	Measure
Evaluated	**Row by row** when data is refreshed	**On the fly**, for the current filter selection
Stored in the model	Yes (uses memory)	No (only the formula)
Result depends on slicers?	No	**Yes**
Use for	New attributes to slice by (Age Band, Hire Year, Profit per row)	Totals, ratios, KPIs, growth
Appears in	Rows, columns, slicers, filters	Values area, cards, tooltips



The same table, two ways. A calculated column adds a value to every row. A measure returns one number computed from whichever rows are currently filtered.

> **TIP** **Rule of thumb**
>
> If you want to **put it on an axis or slicer**, make a column. If you want to **put it in the Values well**, make a measure. In a star schema, almost every number is a measure.

DAX syntax in five minutes

```dax
Total Sales = SUM ( fact_sales[sales_amount] )
```

- `Total Sales` is the **measure name** (spaces allowed, no brackets when defining it).
- `=` starts the formula.
- `SUM` is a **function**; `fact_sales[sales_amount]` is a **fully qualified column** (table name, then column in square brackets). Always include the table name for columns; **never** include it for measures: `[Total Sales]`.
- Strings use double quotes; comments use `--` or `//` or `/* … */`.
- Operators: `+ - * /`, `&` (join text), `= <> > < ≥ ≤`, `&&` (AND), `||` (OR), `IN { "A", "B" }`.
- Spaces and line breaks are ignored; format with **daxformatter.com** or the formatting button.

Creating measures

Home → **New measure** (or right-click a table). Put all measures in an empty table named `_Measures` (Home → Enter data → name `_Measures` → Load, then hide its default column) so they are easy to find. Use **display folders** (Model view → Properties) to group them.

Basic aggregations

```dax
Total Sales      = SUM ( fact_sales[sales_amount] )
Total Cost       = SUM ( fact_sales[total_cost] )
Total Profit     = [Total Sales] - [Total Cost]
Profit Margin %  = DIVIDE ( [Total Profit], [Total Sales] )
Units Sold       = SUM ( fact_sales[quantity] )
Orders           = DISTINCTCOUNT ( fact_sales[order_number] )
Order Lines      = COUNTROWS ( fact_sales )
Avg Order Value  = DIVIDE ( [Total Sales], [Orders] )     -- in this data every order has one line, so Orders = Order Lines
Customers        = DISTINCTCOUNT ( fact_sales[customer_key] )
```

Check the values in Python; they are what a card visual with no filters would display:

Python

```python
import pandas as pd
D = "adventure_cycles/"
f = pd.read_csv(D + "fact_sales.csv", parse_dates=["order_date"]); p = pd.read_csv(D +
"dim_product.csv"); t = pd.read_csv(D + "dim_territory.csv")
sales, cost = f.sales_amount.sum(), f.total_cost.sum()
print(f"Total Sales     : {sales:,.0f}")
print(f"Total Cost      : {cost:,.0f}")
print(f"Total Profit    : {sales - cost:,.0f}")
print(f"Profit Margin % : {(sales - cost) / sales:.1%}")
print(f"Units Sold      : {f.quantity.sum():,}")
print(f"Orders          : {f.order_number.nunique():,}")
print(f"Order Lines     : {len(f):,}")
print(f"Avg Order Value : {sales / f.order_number.nunique():,.0f}")
print(f"Customers       : {f.customer_key.nunique():,}")
```

OUTPUT

```
Total Sales     : 594,559,665
Total Cost      : 384,392,790
Total Profit    : 210,166,875
Profit Margin % : 35.3%
Units Sold      : 22,703
Orders          : 9,000
Order Lines     : 9,000
Avg Order Value : 66,062
Customers       : 1,200
```

WATCH OUT **DIVIDE, not /**

`[A] / [B]` returns an error or infinity when B is 0 or blank. `DIVIDE ( [A], [B] )` returns BLANK (or an alternate result: `DIVIDE ( [A], [B], 0 )`). Always use DIVIDE for ratios.

Filter context: the heart of DAX

When you put a measure in a visual, DAX evaluates it **once per cell**, with a **filter context** made of everything that filters the data for that cell: row and column headers, slicers, page/report filters, and cross-filtering from other visuals.



Filter context for one cell of a matrix (Category = Bikes, Country = India, Year = 2025 slicer). DAX applies all of them to the fact table before it sums.

Same measure, different context in each cell. The matrix below is exactly what Power BI shows for `Total Sales` with Category on rows and Country on columns:

Python

```python
m = f.merge(p, on="product_key").merge(t, on="territory_key")
m["year"] = m.order_date.dt.year
print((m.pivot_table(index="category", columns="country", values="sales_amount",
aggfunc="sum") / 1e6).round(1).to_string())
print("\nSame measure, with a Year = 2025 slicer:")
print((m[m.year == 2025].pivot_table(index="category", columns="country",
values="sales_amount", aggfunc="sum") / 1e6).round(1).to_string())
```

OUTPUT

```
country      India  Singapore   UAE
category
Accessories    6.7        1.7   1.3
Bikes        320.6       74.8  71.3
Clothing       6.0        1.3   1.1
Components    73.5       18.8  17.2

Same measure, with a Year = 2025 slicer:
country      India  Singapore   UAE
category
Accessories    2.2        0.7   0.4
Bikes        102.3       24.6  22.0
Clothing       2.0        0.3   0.4
Components    22.9        7.1   5.1
```

Row context

A **row context** exists when DAX goes through a table row by row: in a **calculated column**, and inside **iterator** functions (`SUMX` , `AVERAGEX` , `FILTER` ...). In row context, column references mean "this row's value".

DAX

```
-- Calculated column on fact_sales: row context gives each row's own values
Line Profit = fact_sales[sales_amount] - fact_sales[total_cost]
```

Row context does **not** automatically filter other tables; **CALCULATE** converts a row context into a filter context ("context transition").

Iterators: SUMX, AVERAGEX, MAXX

`SUM ( column )` adds one column. When you need to **calculate per row first** and then aggregate, use an iterator: `SUMX ( table, expression )`.

DAX

```
-- Revenue at list price (unit_price x quantity), calculated row by row
List Value = SUMX ( fact_sales, fact_sales[quantity] * fact_sales[unit_price] )

-- Average margin per order line (not the same as total profit / total sales!)
Avg Line Margin % = AVERAGEX ( fact_sales, DIVIDE ( fact_sales[sales_amount] -
fact_sales[total_cost], fact_sales[sales_amount] ) )

-- Most profitable single line
Max Line Profit = MAXX ( fact_sales, fact_sales[sales_amount] - fact_sales[total_cost] )
```

Python

```
f["line_profit"] = f.sales_amount - f.total_cost
list_value = (f.quantity * f.unit_price).sum()
avg_line_margin = ((f.sales_amount - f.total_cost) / f.sales_amount).mean()
print(f"List Value        : {list_value:,.0f}")
print(f"Realised/List     : {f.sales_amount.sum() / list_value:.1%}  (average price
realisation)")
print(f"Avg Line Margin % : {avg_line_margin:.1%}   vs overall margin
{(f.sales_amount.sum() - f.total_cost.sum()) / f.sales_amount.sum():.1%}")
print(f"Max Line Profit   : {f.line_profit.max():,.0f}")
```

OUTPUT

```
List Value        : 617,201,300
Realised/List     : 96.3%  (average price realisation)
Avg Line Margin % : 34.5%   vs overall margin 35.3%
Max Line Profit   : 174,060
```

The average of line margins (34.5%) differs slightly from the overall margin (35.3%) because large lines count more in the total than in a simple average. Choose the definition the business wants, and

name the measure clearly.

CALCULATE: change the filter context

`CALCULATE ( expression, filter1, filter2, … )` evaluates the expression in a **modified** filter context. It is the most important DAX function.

DAX

```
Bike Sales   = CALCULATE ( [Total Sales], dim_product[category] = "Bikes" )
India Sales  = CALCULATE ( [Total Sales], dim_territory[country] = "India" )
Bike Sales India 2025 =
    CALCULATE ( [Total Sales], dim_product[category] = "Bikes", dim_territory[country] =
"India", dim_date[year] = 2025 )
High Value Lines = CALCULATE ( COUNTROWS ( fact_sales ), fact_sales[sales_amount] > 200000
)
```

Multiple filter arguments are combined with **AND**. A simple column filter like `dim_product[category] = "Bikes"` **replaces** any existing filter on that column.

Python

```
bike = f.merge(p, on="product_key").query("category == 'Bikes'").sales_amount.sum()
print(f"Bike Sales          : {bike:,.0f}")
india_2025_bikes = m[(m.category == "Bikes") & (m.country == "India") & (m.year ==
2025)].sales_amount.sum()
print(f"Bike Sales India 2025: {india_2025_bikes:,.0f}")
print(f"High Value Lines    : {(f.sales_amount > 200000).sum():,}")
```

OUTPUT

```
Bike Sales          : 466,754,465
Bike Sales India 2025: 102,271,520
High Value Lines    : 797
```

Filter functions: ALL, REMOVEFILTERS, ALLSELECTED, ALLEXCEPT, KEEPFILTERS

Function	Effect	Typical use
`ALL ( table or column )` / `REMOVEFILTERS ( … )`	Ignores filters on the table/column	Grand totals, denominators
`ALLSELECTED ( … )`	Ignores filters from the visual's own rows/columns but keeps slicers and page filters	"% of visible total"
`ALLEXCEPT ( table, col1, … )`	Removes all filters on the table except those columns	Subtotals per group
`VALUES ( column )`	The distinct values visible in the current context	Counting / looping over what is filtered
`SELECTEDVALUE ( column, alt )`	The single selected value (or alternate)	Titles, dynamic measures
`KEEPFILTERS ( filter )`	Intersects with, rather than replaces, existing filters	Advanced

Percent of total

DAX

```
Sales % of Total =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], REMOVEFILTERS ( dim_product ) ) )

Sales % of Visible Total =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], ALLSELECTED ( dim_product ) ) )

Share Within Country =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], ALLEXCEPT ( dim_territory,
dim_territory[country] ) ) )
```

Python

```
cat = m.groupby("category").sales_amount.sum()
print((100 * cat / cat.sum()).round(1).rename("Sales % of Total").to_string())
```

OUTPUT

```
category
Accessories     1.6
Bikes          78.5
```

```
Clothing        1.4
Components     18.4
```

FILTER: row-by-row conditions

`FILTER ( table, condition )` is an iterator that returns the rows where the condition is TRUE. Use it when a filter needs a **measure or several columns**; for a simple single-column comparison, the plain `column = value` form inside CALCULATE is faster.

DAX

```dax
-- Sales from products with a list price above 1,00,000
Premium Sales = CALCULATE ( [Total Sales], FILTER ( dim_product, dim_product[list_price] > 100000 ) )

-- Number of products whose total sales beat 1 crore
Strong Products = COUNTROWS ( FILTER ( VALUES ( dim_product[product] ), [Total Sales] > 10000000 ) )
```

Python

```python
prem = f.merge(p, on="product_key").query("list_price > 100000").sales_amount.sum()
ps = f.groupby("product_key").sales_amount.sum()
print(f"Premium Sales  : {prem:,.0f}")
print(f"Strong Products: {(ps > 10_000_000).sum()}  (products with sales above ₹1 crore)")
```

OUTPUT

```
Premium Sales  : 303,639,995
Strong Products: 20  (products with sales above ₹1 crore)
```

Variables and readability

`VAR … RETURN` stores intermediate results: easier to read, easier to debug, and faster (each variable is evaluated once).

DAX

```dax
Profit Margin vs Overall =
VAR CurrentMargin = DIVIDE ( [Total Profit], [Total Sales] )
VAR OverallMargin = CALCULATE ( DIVIDE ( [Total Profit], [Total Sales] ), REMOVEFILTERS () )
RETURN
    CurrentMargin - OverallMargin
```

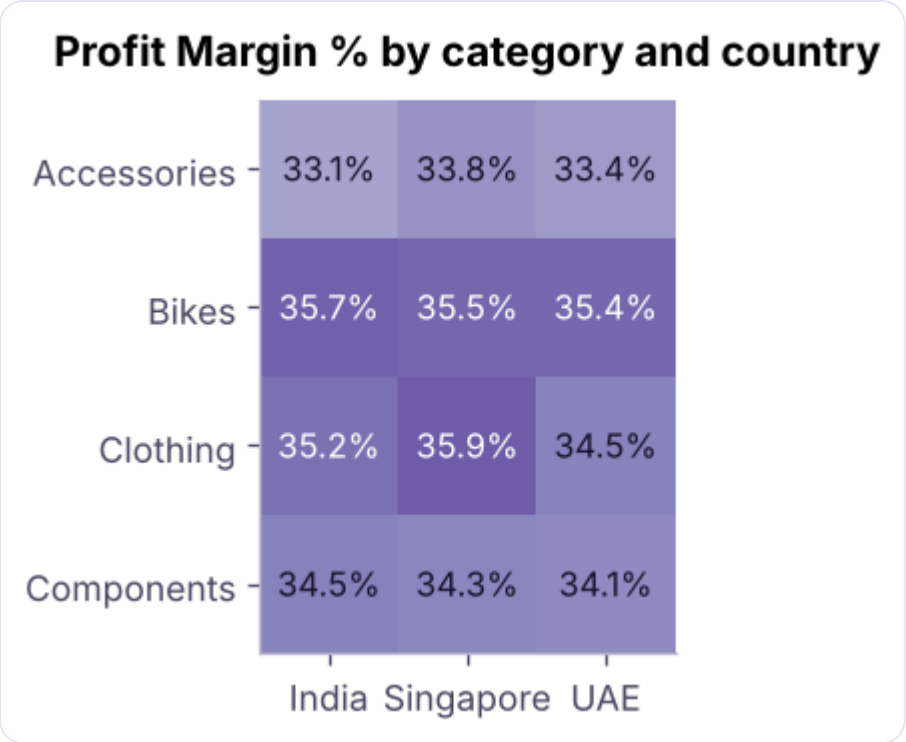
Logical and text measures

```dax
Margin Status =
SWITCH (
    TRUE (),
    ISBLANK ( [Profit Margin %] ), "No data",
    [Profit Margin %] ⩾ 0.4,      "Excellent",
    [Profit Margin %] ⩾ 0.3,      "Good",
    "Review"
)

Selected Country Title =
"Sales for " & SELECTEDVALUE ( dim_territory[country], "all countries" )
```

Function	Purpose
`IF ( test, a, b )`, `SWITCH ( expr, v1, r1, … )`, `SWITCH ( TRUE (), cond1, r1, … )`	Branching
`ISBLANK`, `COALESCE ( x, 0 )`, `BLANK ()`	Handle empty values
`HASONE ( col )`, `ISFILTERED ( col )`	Test the current context
`FORMAT ( value, "0.0%" )`, `CONCATENATEX ( table, col, ", " )`	Text

Measures in action: a profit-margin matrix



Profit margin % by category and country: the matrix you would build with the Profit Margin % measure and conditional formatting (darker = higher).

Common DAX mistakes

Mistake	Fix
Using a calculated column where a measure is needed (results do not respond to slicers)	Create a measure
Dividing with `/`	`DIVIDE`
Writing `SUM ( [Measure] )`	`SUM` takes columns; use `SUMX ( table, [Measure] )` for measures over rows
Averaging ratios (`AVERAGE` of a margin column)	Compute `DIVIDE ( SUM numerator, SUM denominator )` unless you really want the average of ratios
Using `FILTER ( ALL ( table ), … )` when a column filter works	Prefer simple column filters inside CALCULATE
Forgetting that a measure's result depends on context; testing in a card	Test in a table or matrix with the fields you care about
Hard-coding values (`"Bikes"`) in many measures	Use a parameter table or SELECTEDVALUE

DO IT WITH AI

Do it with AI

Copilot and chat assistants write decent DAX, but they invent column names and often misuse context. A reliable prompt gives the **exact table and column names**, the **relationships**, and the **expected result**: *"Tables: fact_sales(order_date, product_key, sales_amount, total_cost), dim_product(product_key, category), dim_date(date). Write a DAX measure for each category's share of total sales that ignores the category filter but respects year slicers. Explain the filter context."* Then test the measure in a table next to your own pandas check.

Summary

KEY TAKEAWAYS

- **Calculated columns** run row by row at refresh and are stored; **measures** run at query time in the current **filter context**.
- Filter context = row/column headers + slicers + filters + cross-filtering; measures are re-evaluated for every cell.
- Row context exists in calculated columns and iterators (`SUMX` , `AVERAGEX` , `FILTER`); `CALCULATE` converts it to filter context.
- `CALCULATE ( expression, filters )` changes the filters; `REMOVEFILTERS` / `ALL` , `ALLSELECTED` , `ALLEXCEPT` remove them for percent-of-total calculations.
- Use `DIVIDE` , variables (`VAR … RETURN`), `SWITCH ( TRUE (), … )` , clear names, a `_Measures` table and display folders.

Quick quiz

6 questions · tap an answer to check it

1. Which is evaluated for every cell of a matrix based on slicers and row/column headers?

- **A** A calculated column
- **B** A measure
- **C** A relationship
- **D** A parameter

Answer: B. Measures respond to filter context; calculated columns are fixed at refresh.

2. Which function lets you compute a total that ignores the Category filter (for a denominator)?

- **A** `SUM`
- **B** `CALCULATE ( [Total Sales], REMOVEFILTERS ( dim_product[category] ) )`
- **C** `FILTER ( [Total Sales] )`
- **D** `COUNTROWS`

Answer: B. REMOVEFILTERS (or ALL) clears the filter inside CALCULATE.

3. Which is the safest way to compute a margin?

- **A** `[Profit] / [Sales]`
- **B** `DIVIDE ( [Profit], [Sales] )`
- **C** `IF ( [Sales] = 0, 0, [Profit] ) / [Sales]`
- **D** `SUM ( [Profit] ) / SUM ( [Sales] )`

Answer: B. DIVIDE handles zero and blank denominators gracefully.

4. `SUMX ( fact_sales, fact_sales[quantity] * fact_sales[unit_price] )` **is needed because:**

- **A** SUM cannot add numbers
- **B** The product must be calculated per row before adding
- **C** SUMX is faster than SUM
- **D** It creates a calculated column

Answer: B. SUM works on one existing column; SUMX evaluates an expression per row.

5. What is row context?

- **A** The current row being evaluated in a calculated column or iterator
- **B** The filters from slicers
- **C** The relationship direction
- **D** A table filter

Answer: A. Row context refers to "this row" during row-by-row evaluation.

6. What does `CALCULATE ( [Total Sales], dim_product[category] = "Bikes" )` **return in a visual that shows each country?**

- **A** Total sales for all products
- **B** Bike sales for each country in the visual
- **C** Bike sales for all countries on every row
- **D** An error

Answer: B. The Category filter is replaced by Bikes, while the Country filter from the visual still applies.

EXERCISE **Write the measures**

Using the Adventure Cycles model, write DAX for: (1) Profit and margin % for **Clothing** only; (2) each country's share of total sales (works in a visual with country on rows); (3) the number of customers who bought more than one product line; (4) a title measure that says "Top category: ..." for the category with the highest sales in the current context.

Show solution (worked answer)

DAX

```
-- (1)
Clothing Profit  = CALCULATE ( [Total Profit], dim_product[category] = "Clothing" )
Clothing Margin  = DIVIDE ( [Clothing Profit], CALCULATE ( [Total Sales],
dim_product[category] = "Clothing" ) )
-- (2)
Country Share = DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], REMOVEFILTERS (
dim_territory ) ) )
-- (3)
Multi-line Customers =
COUNTROWS ( FILTER ( VALUES ( fact_sales[customer_key] ), CALCULATE ( COUNTROWS (
fact_sales ) ) > 1 ) )
-- (4)
Top Category =
VAR t = ADDCOLUMNS ( VALUES ( dim_product[category] ), "@s", [Total Sales] )
VAR top1 = TOPN ( 1, t, [@s], DESC )
RETURN "Top category: " & MAXX ( top1, dim_product[category] )
```

Verify each in a table visual next to a pandas check.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between a calculated column and a measure? A column is computed per row at refresh and stored; a measure is computed at query time for the current filter context and returns a single value.

Q: Explain filter context and row context. Filter context is the set of filters (axes, slicers, filters) applied to the model when a measure is evaluated. Row context is the "current row" while iterating a table (calculated columns, iterators). CALCULATE turns row context into filter context.

Q: What does CALCULATE do? Evaluates an expression in a modified filter context; its filter arguments add, replace or remove filters.

Q: ALL vs ALLSELECTED? ALL removes all filters from the specified table/column (grand total); ALLSELECTED removes only the filters coming from the visual's own rows and columns but keeps outer filters such as slicers, giving a total of what is visible.

Q: SUM vs SUMX? SUM adds a column; SUMX evaluates an expression for each row of a table and sums the results (needed for products of columns, conditional logic per row).

POWER BI · LESSON 4

DAX Time Intelligence, Ranking, Parameters & Dynamic Measures

Make dashboards answer "compared with when?". Build year-to-date, month-to-date, same-period-last-year, month-on-month and year-on-year measures, rolling averages, rankings with RANKX and TOPN, what-if parameters, field parameters and dynamic measure switching, and know visual calculations and calculation groups.

● Power BI classes 3–4

✦ Power BI Desktop 2026 (field parameters, calculation groups, visual calculations)

Advanced

⏱ 150 min

By the end of this lesson you can

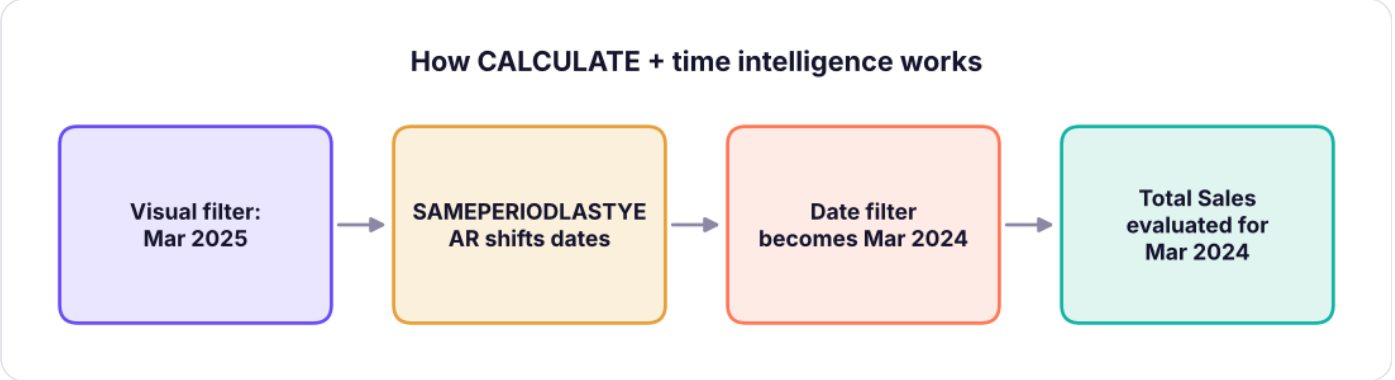
- Explain what time intelligence needs (a marked date table) and use TOTALYTD, DATESYTD, SAMEPERIODLASTYEAR, DATEADD and PARALLELPERIOD
- Build YoY %, MoM %, YTD, rolling 3-month and running-total measures
- Rank items with RANKX and show the Top N with TOPN, including an "Others" bucket
- Create what-if parameters and dynamic measure switching with field parameters
- Describe calculation groups, visual calculations and DAX query view, and when to use them

What time intelligence needs

Time-intelligence functions move or expand the **date filter**: "this year to date", "the same period last year". They only work when:

1. You have a **date table** with every date (Lesson 2),
2. it is **marked as a date table**, and
3. it is **related** to the fact table's date column.

Always use the **date table's** `Date` column inside time functions, never the fact table's date.



Time intelligence changes the date filter, then CALCULATE re-evaluates the measure. SAMEPERIODLASTYEAR shifts the selected dates back exactly one year.

Core measures

DAX

```
Total Sales = SUM ( fact_sales[sales_amount] )

-- Cumulative totals
Sales YTD = TOTALYTD ( [Total Sales], dim_date[date] )
Sales QTD = TOTALQTD ( [Total Sales], dim_date[date] )
Sales MTD = TOTALMTD ( [Total Sales], dim_date[date] )
-- Indian financial year ending 31 March:
Sales FYTD = TOTALYTD ( [Total Sales], dim_date[date], "31-3" )

-- Previous periods
Sales LY    = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( dim_date[date] ) )
Sales PM    = CALCULATE ( [Total Sales], DATEADD ( dim_date[date], -1, MONTH ) )
Sales PQ    = CALCULATE ( [Total Sales], DATEADD ( dim_date[date], -1, QUARTER ) )

-- Growth
YoY %  = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )
MoM %  = DIVIDE ( [Total Sales] - [Sales PM], [Sales PM] )
YoY Δ  = [Total Sales] - [Sales LY]
```

Function	Returns	Notes
`TOTALYTD ( expr, dates, [year_end] )`	Year-to-date total	Shortcut for `CALCULATE ( expr, DATESYTD ( dates ) )`
`DATESYTD / DATESQTD / DATESMTD`	The date set from the period start to the last visible date	Use inside CALCULATE
`SAMEPERIODLASTYEAR ( dates )`	The selected dates shifted back one year	Needs continuous dates
`DATEADD ( dates, n, interval )`	Shift by n days, months, quarters or years	Month/quarter shifts handle month-end well
`PARALLELPERIOD ( dates, n, interval )`	The **full** period shifted (full month, quarter, year)	
`DATESBETWEEN ( dates, start, end )`	Custom ranges	
`DATESINPERIOD ( dates, end, -n, interval )`	Rolling window ending at `end`	Rolling 3/12 months
`PREVIOUSMONTH / PREVIOUSQUARTER / PREVIOUSYEAR`	Whole previous period	
`EOMONTH`, `EDATE`, `DATEDIFF`	Date arithmetic	

Let's compute what these measures return for our data (monthly sales, year-to-date, last year and YoY):

Python

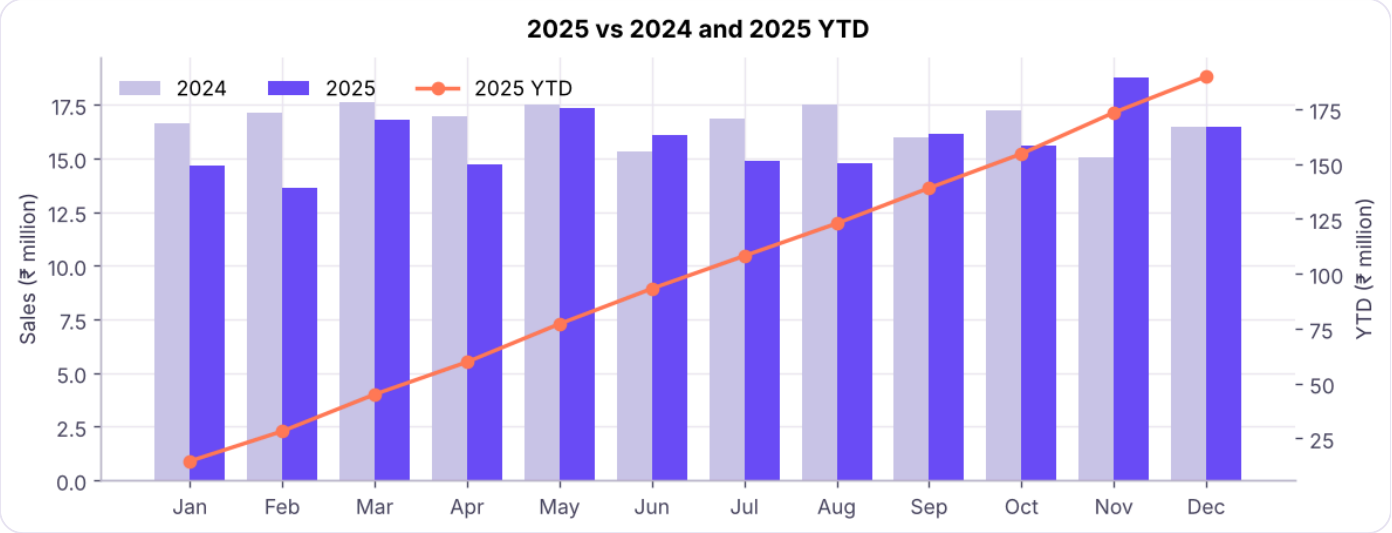
```
import pandas as pd
D = "adventure_cycles/"
f = pd.read_csv(D + "fact_sales.csv", parse_dates=["order_date"]); p = pd.read_csv(D +
"dim_product.csv")
f["ym"] = f.order_date.dt.to_period("M")
mth = f.groupby("ym").sales_amount.sum().to_frame("Total Sales")
mth["Sales PM"] = mth["Total Sales"].shift(1)
mth["MoM %"] = (mth["Total Sales"] / mth["Sales PM"] - 1) * 100
mth["Sales LY"] = mth["Total Sales"].shift(12)
mth["YoY %"] = (mth["Total Sales"] / mth["Sales LY"] - 1) * 100
mth["Sales YTD"] = mth.groupby(mth.index.year)["Total Sales"].cumsum()
out = mth[mth.index.year == 2025].copy()
out["Total Sales"] = (out["Total Sales"] / 1e6).round(1); out["Sales LY"] = (out["Sales
LY"] / 1e6).round(1); out["Sales YTD"] = (out["Sales YTD"] / 1e6).round(1)
out[["Total Sales", "Sales LY", "YoY %", "MoM %", "Sales YTD"]].round(1).rename_axis("2025
month")
```

OUTPUT

	Total Sales	Sales LY	YoY %	MoM %	Sales YTD
2025 month					
2025-01	14.6	16.7	-12.1	-11.2	14.6
2025-02	13.6	17.1	-20.5	-6.9	28.3
2025-03	16.8	17.6	-4.8	23.1	45.1
2025-04	14.7	17.0	-13.2	-12.1	59.8
2025-05	17.4	17.5	-1.0	17.7	77.2
2025-06	16.1	15.3	5.0	-7.4	93.2
2025-07	14.9	16.9	-11.5	-7.2	108.1
2025-08	14.8	17.5	-15.6	-1.0	122.9
2025-09	16.2	16.0	1.0	9.4	139.1
2025-10	15.6	17.3	-9.7	-3.5	154.7
2025-11	18.8	15.1	24.6	20.5	173.4
2025-12	16.5	16.5	0.0	-12.2	189.9

12 rows × 5 columns

(Values in ₹ million.) In a Power BI matrix with `dim_date[month]` on rows and these measures in the Values area you would see exactly this.



Monthly sales for 2025 against the same month in 2024, with year-to-date sales on the right axis. This is the classic "this year vs last year" chart.

COMMON MISTAKES

Common time-intelligence bugs

- **Date table not marked / has gaps**: `SAMEPERIODLASTYEAR` returns blank or wrong values.
- **Using the fact table's date column** in the function: use `dim_date[date]`.
- **Comparing partial periods**: the current month has data for 10 days, last year's month has 31, so YoY looks terrible. Compare "to date" with `DATESYTD` and `SAMEPERIODLASTYEAR`, or restrict to complete months.
- **Fiscal year**: pass `"31-3"` as the year-end argument to `TOTALYTD`, and create `Fiscal Year` and `Fiscal Quarter` columns in the date table.
- **Showing a "last year" value in months with no data** (future months): wrap with `IF ( [Total Sales] <> BLANK (), … )`.

Rolling windows and running totals

DAX

```
-- Rolling 3-month sales ending at the last visible date
Sales Rolling 3M =
CALCULATE ( [Total Sales], DATESINPERIOD ( dim_date[date], MAX ( dim_date[date] ), -3,
MONTH ) )

-- Rolling 3-month average of monthly sales (average over the months in the window)
Avg Monthly Sales R3M =
DIVIDE ( [Sales Rolling 3M], 3 )      -- valid once three full months exist in the window

-- Running total across all dates up to the current one
Sales Running Total =
CALCULATE ( [Total Sales], FILTER ( ALL ( dim_date[date] ), dim_date[date] ≤ MAX (
dim_date[date] ) ) )
```

In SQL this was `SUM() OVER (ORDER BY … ROWS BETWEEN …)` (SQL Lesson 12). DAX does the same thing by redefining the date filter.

Python

```
mth["Rolling 3M"] = mth["Total Sales"].rolling(3).sum()
mth["Rolling 3M avg"] = mth["Total Sales"].rolling(3).mean()
(mth[["Total Sales", "Rolling 3M", "Rolling 3M avg"]].tail(6) /
1e6).round(1).rename_axis("month (₹ million)")
```

OUTPUT

month (₹ million)	Total Sales	Rolling 3M	Rolling 3M avg
2025-07	14.9	48.3	16.1
2025-08	14.8	45.8	15.3
2025-09	16.2	45.8	15.3
2025-10	15.6	46.5	15.5
2025-11	18.8	50.5	16.8
2025-12	16.5	50.9	17.0

6 rows × 3 columns

Ranking: RANKX and TOPN

DAX

```
-- Rank of each product category by sales within the current context
Category Rank = RANKX ( ALL ( dim_product[category] ), [Total Sales], , DESC, DENSE )

-- Show only the top 5 products: sales of a product if it is in the top 5, else blank
Top 5 Product Sales =
VAR top5 = TOPN ( 5, ALL ( dim_product[product] ), [Total Sales], DESC )
RETURN CALCULATE ( [Total Sales], KEEPFILTERS ( top5 ) )

-- Everything outside the top 5, as one "Others" value
Others Sales = [Total Sales] - [Top 5 Product Sales]
```

For simple Top N visuals, you do not need DAX: in the **Filters pane**, choose the visual's field → Filter type **Top N** → Show items Top 5 → By value `Total Sales` → Apply filter. Use DAX when the Top N must respond to slicers in a custom way or you need an "Others" bar.

Python

```
prod_sales = f.merge(p, on="product_key").groupby(["category",
"product"]).sales_amount.sum().sort_values(ascending=False)
top5 = prod_sales.head(5)
print((top5 / 1e6).round(1).to_string())
print(f"\nTop 5 total : {top5.sum() / 1e6:,.1f} M   |   Others: {(prod_sales.sum() -
top5.sum()) / 1e6:,.1f} M   |   Top 5 share: {top5.sum() / prod_sales.sum():.1%}")
```

OUTPUT

```
category  product
Bikes     Road Bike E-331        51.3
          Mountain Bike C-245    42.4
          Touring Bike A-507     39.7
          Mountain Bike D-662    36.2
          Mountain Bike B-168    35.9

Top 5 total : 205.4 M   |   Others: 389.2 M   |   Top 5 share: 34.5%
```

Parameters

What-if parameter

Modeling → New parameter → Numeric range: name `Price Change %`, from −20 to +20, increment 1, default 0, and tick **Add slicer**. Power BI creates a table and a measure `Price Change % Value`. Use it in your own measure to simulate:

DAX

```
Simulated Sales = [Total Sales] * ( 1 + [Price Change % Value] / 100 )
Simulated Profit = [Simulated Sales] - [Total Cost]
```

Dragging the slicer instantly shows what profit would be with a 5% price increase (ignoring demand changes: always say so on the page).

Python

```
base_sales, base_cost = f.sales_amount.sum(), f.total_cost.sum()
rows = [(chg, base_sales * (1 + chg / 100) / 1e6, (base_sales * (1 + chg / 100) -
base_cost) / 1e6) for chg in (-10, -5, 0, 5, 10)]
pd.DataFrame(rows, columns=["Price change %", "Simulated sales (₹ M)", "Simulated profit (₹
M)"]).round(1)
```

OUTPUT

	Price change %	Simulated sales (₹ M)	Simulated profit (₹ M)
0	-10	535.1	150.7
1	-5	564.8	180.4
2	0	594.6	210.2
3	5	624.3	239.9
4	10	654.0	269.6

5 rows × 3 columns

Power Query parameters

Home → Manage parameters creates parameters used inside queries, for example `StartDate` or a `ServerName`. Use them to switch between a test and a production database, or to filter large tables at load time. In the Service, parameters can be changed in dataset settings.

Dynamic measure switching

A very popular pattern: a slicer that lets users choose which measure the chart shows (Sales, Profit, Orders).

The modern way: Field parameters. Modeling → **New parameter → Fields**: add `Total Sales`, `Total Profit`, `Orders`; tick **Add slicer**. Power BI creates a table with the three measures; use that

table's field on a chart's axis or values and the slicer swaps the measure. A field parameter can also switch **axis** fields (Category, Country, Year).

The classic way: a disconnected table + SWITCH.

```DAX
-- Table "Measure Selector" (Enter data): columns Measure Name, Sort
Selected Measure =
SWITCH (
    SELECTEDVALUE ( 'Measure Selector'[Measure Name], "Sales" ),
    "Sales",  [Total Sales],
    "Profit", [Total Profit],
    "Orders", [Orders]
)
```

Format strings can be dynamic too (Power BI 2024+ "dynamic format string"): thousands for sales, percentage for margin.

KPI visuals and forecasting

- **KPI visual**: needs a value (indicator), a trend axis (date) and a target (for example last year's sales). Colours show good/bad.
- **Gauge**: current vs target; min/max/target set in the format pane.
- **Forecasting**: in a **line chart** with a date axis, Analytics pane → **Forecast** → set forecast length (for example 6 months), confidence interval (95%) and seasonality (12 points). Power BI uses exponential smoothing (ETS). It is a quick what-comes-next view, not a replacement for proper forecasting models (ML track).
- **Analytics pane** also offers constant lines, average, min/max, percentile lines, trend lines, and **anomaly detection**.

Newer DAX features worth knowing

Feature	What it is
Calculation groups	A reusable "time-intelligence switcher": define `Current`, `YTD`, `LY`, `YoY %` once and apply them to every measure (created in Tabular Editor or in the model view in recent versions)
Visual calculations	DAX written **inside a visual** that can refer to other fields in the visual (running sum, moving average, % of parent) with simple functions such as `RUNNINGSUM`, `MOVINGAVERAGE`, `PREVIOUS`, `FIRST`
DAX query view	`EVALUATE` queries to test and debug measures; Copilot can write DAX here
User-defined functions (DAX UDFs)	Reusable functions defined with the `FUNCTION` keyword (preview)
INFO.* functions	Query model metadata (`INFO.TABLES()`, `INFO.MEASURES()`)

DAX

```dax
-- Visual calculation (typed in the visual's formula bar)
Running Sales = RUNNINGSUM ( [Total Sales] )
Moving Avg 3  = MOVINGAVERAGE ( [Total Sales], 3 )
```

DO IT WITH AI

Do it with AI

Copilot can write time-intelligence measures from a sentence ("Show sales year to date versus last year"). It works best if the date table is marked, columns and tables have clear names, and you add **descriptions** to measures (Model view → Properties). Check the generated DAX uses your `dim_date[date]` (not the fact table's date), and confirm one value by hand.

Summary

KEY TAKEAWAYS

- Time intelligence needs a **continuous, marked date table**, related to the fact; always reference the date table's column.
- `TOTALYTD`, `SAMEPERIODLASTYEAR`, `DATEADD`, `PARALLELPERIOD`, `DATESINPERIOD` build YTD, last-year, MoM, YoY and rolling windows; `DIVIDE ( current - previous, previous )` gives growth.
- `RANKX` ranks, `TOPN` selects the top items (add an "Others" measure); the Filters-pane **Top N** is the no-code route.
- **What-if parameters** simulate scenarios; **field parameters** let users switch measures or axes; the disconnected table + `SWITCH` pattern does the same in older models.
- Know calculation groups, visual calculations and the DAX query view for advanced work.

Quick quiz

6 questions · tap an answer to check it

1. **Which measure shows year-to-date sales?**

A `SUM ( fact_sales[sales_amount] )`

B `TOTALYTD ( [Total Sales], dim_date[date] )`

C `SAMEPERIODLASTYEAR ( [Total Sales] )`

D `RANKX ( dim_date, [Total Sales] )`

Answer: B. TOTALYTD accumulates from the start of the year to the current date.

2. SAMEPERIODLASTYEAR returns blank for every month. What should you check first?

A The theme

B That the date table is continuous, marked as a date table and related to the fact

C The colour of the visual

D The number of slicers

Answer: B. Time intelligence relies on a valid date table.

3. Which formula gives month-on-month growth?

A `[Total Sales] - [Sales PM]`

B `DIVIDE ( [Total Sales] - [Sales PM], [Sales PM] )`

C `SUM ( [Sales PM] ) / [Total Sales]`

D `DATEADD ( [Total Sales], -1, MONTH )`

Answer: B. Growth rate = (current − previous) ÷ previous, using DIVIDE.

4. What does a field parameter let users do?

A Change the report theme

B Switch which measures or fields a visual shows with a slicer

C Refresh the data

D Create relationships

Answer: B. Field parameters create a slicer that swaps fields in visuals.

5. In the Indian financial year (April–March), how do you build YTD sales?

A `TOTALYTD ( [Total Sales], dim_date[date] )` always works

B `TOTALYTD ( [Total Sales], dim_date[date], "31-3" )`

C `DATESMTD`

D `PARALLELPERIOD ( … )`

Answer: B. The third argument sets the year-end date.

6. What is the correct way to compute "Others" after a Top 5?

A Add a second visual

B Total sales minus the Top 5 measure

C Use FILTER ALL

D Use PREVIOUSMONTH

Answer: B. Others = overall total − Top N total.

EXERCISE

Time-intelligence measures

For Adventure Cycles write measures for: (1) sales for the last 12 months ending at the selected date; (2) YoY % by category; (3) the best month of each year; (4) a what-if parameter "Discount %" from 0 to 30 that reduces `Total Sales` and shows simulated profit margin.

Show solution (worked answer)

DAX

```
-- (1)
Sales L12M = CALCULATE ( [Total Sales], DATESINPERIOD ( dim_date[date], MAX ( dim_date[date] ), -12, MONTH ) )
-- (2) put dim_product[category] on rows and dim_date[year] on columns
YoY % = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )
-- (3)
Best Month =
VAR t = ADDCOLUMNS ( VALUES ( dim_date[month] ), "@s", [Total Sales] )
RETURN MAXX ( TOPN ( 1, t, [@s], DESC ), dim_date[month] )
-- (4)  parameter table 'Discount %' with measure [Discount % Value]
Simulated Margin = DIVIDE ( [Total Sales] * ( 1 - [Discount % Value] / 100 ) - [Total Cost], [Total Sales] * ( 1 - [Discount % Value] / 100 ) )
```

Validate (2) and (3) against pandas group-bys.

INTERVIEW CORNER

Questions you may be asked

Q: What do you need for time-intelligence functions to work? A date table with continuous dates covering the data, marked as a date table, related to the fact table by date.

Q: How would you calculate YoY growth? `Sales LY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( dim_date[date] ) )` then `YoY % = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )`.

Q: How can users choose which metric a chart shows? Field parameters (modern) or a disconnected table with a `SWITCH ( SELECTEDVALUE (…), … )` measure.

Q: How do you show Top N plus an "Others" bar? A Top N measure using TOPN/KEEPFILTERS and an Others measure = total − Top N.

POWER BI · LESSON 5

Visuals, Filters, Drill-through, Tooltips, Buttons & Report Design

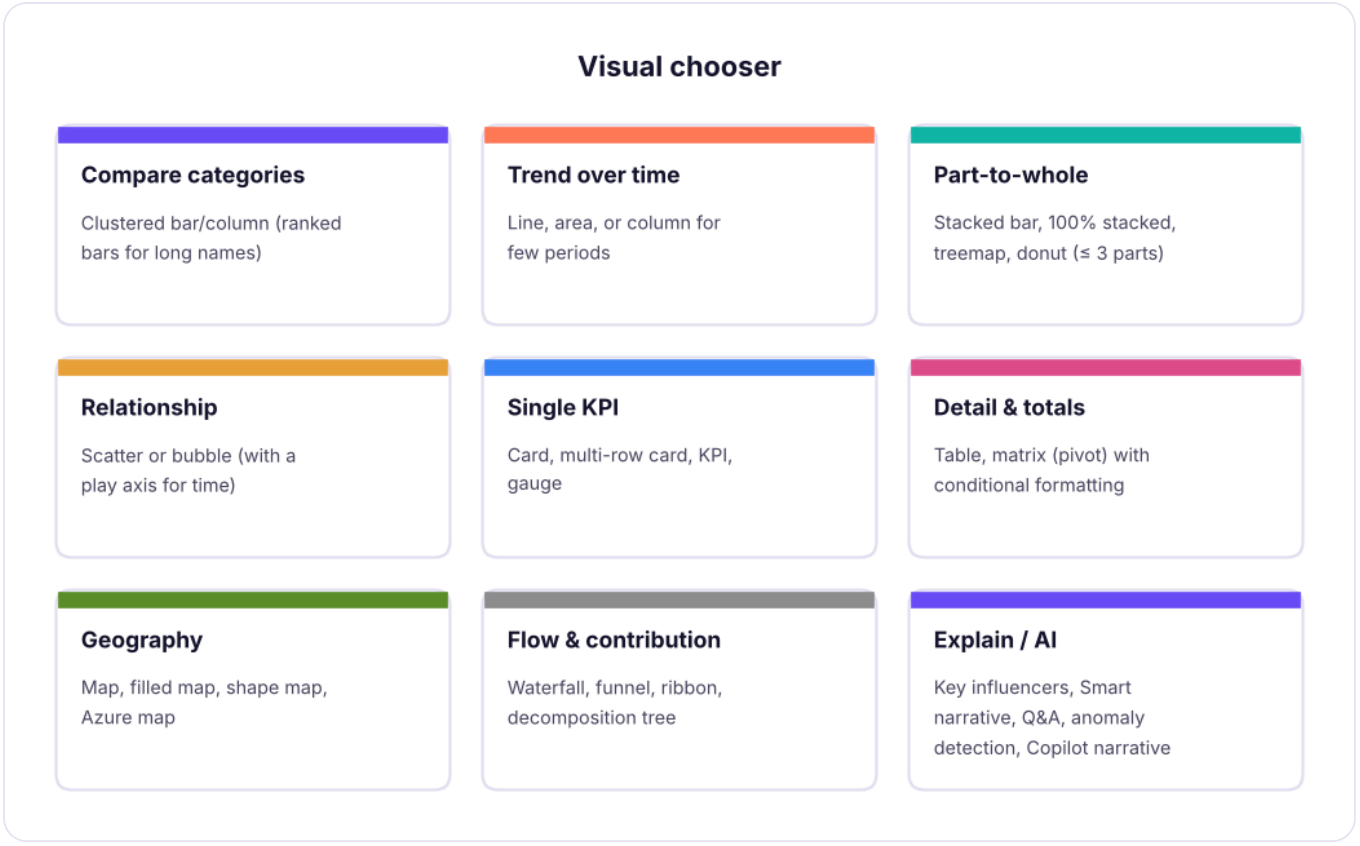
Turn a model into an interactive, well-designed report. Choose visuals, format them (cards, matrix, charts, gauge, KPI, maps, images), use visual, page and report filters, edit interactions, drill-down and drill-through, custom tooltips, bookmarks and buttons, conditional formatting, themes and design principles.

Power BI classes 2–4 | Power BI Desktop 2026 | Intermediate | 150 min

By the end of this lesson you can

- Pick the right visual for each question and configure its field wells and formatting
- Apply visual, page and report-level filters, Top N, relative date filters and slicer sync
- Control cross-filtering with Edit interactions; add drill-down, drill-through and report-page tooltips
- Build navigation with buttons, bookmarks, page navigation and a selection pane
- Apply themes and conditional formatting, and follow layout, colour and accessibility rules

Choosing the visual



Match the visual to the question.

Visual	Fields	Good for	Watch out
Card / multi-row card	Values	One or few headline numbers	Format display units (K, M, Cr) and add a clear label
Clustered bar / column	Axis, Values, Legend	Comparisons	Sort descending, start at zero, ≤ 15 bars
Stacked / 100% stacked	Axis, Legend, Values	Composition	Only the bottom segment is easy to compare
Line / area	X-axis (date), Y-axis, Legend	Trends	Limit to ~5 lines; use a continuous date axis
Line + clustered column	Shared axis, column values, line values	Two measures with different scales (secondary axis)	Label both axes
Waterfall	Category, Y-axis	Bridge from start to end (profit walk)	Needs an ordered category
Scatter / bubble	X, Y, Size, Details, Play axis	Relationships and outliers	Add Details so each point is an entity
Treemap	Group, Values	Hierarchical share of total	Many tiny rectangles are unreadable
Donut / pie	Legend, Values	2–3 parts	More than 4 slices = use a bar
Matrix	Rows, Columns, Values	Pivot tables with drill and subtotals	Use conditional formatting for heat maps
Table	Columns	Exact values, detail list	Keep it short; a table is not a dashboard
KPI / **Gauge**	Value, Target, Trend axis	Progress to a target	Gauges waste space; prefer KPI cards
Funnel	Stage, Values	Drop-off between stages	
Map / Filled map	Location, Size, Legend	Geography	Geocoding needs clean place names (`City, State, Country`)
Slicer	Field	Filter the page	Styles: list, dropdown, tiles, between (range), relative date
Decomposition tree	Analyze, Explain by	Drill into drivers	Interactive exploration

Visual	Fields	Good for	Watch out
Key influencers	Analyze, Explain by	AI finds what drives a metric	Needs a binary/numeric target
Smart narrative	Auto-text	Written summary that updates	Check the wording

Formatting essentials

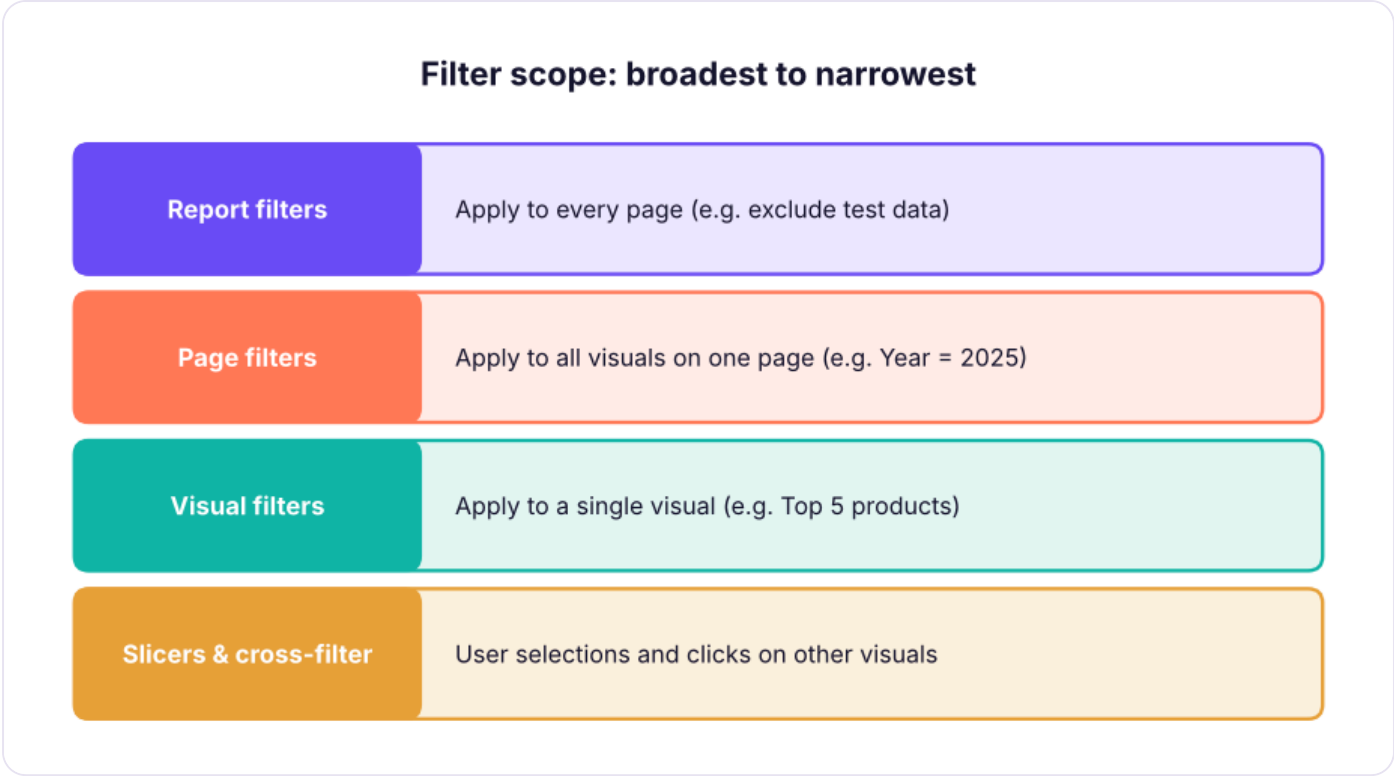
Select a visual → **Format visual** (paint-roller icon): **Visual** tab (axis, labels, colours, legend, gridlines, title) and **General** (title, size, position, background, border, alt text).

Task	Where
Display units and decimals	Format → Callout value / Data labels → Display units: Thousands, Lakhs (not built in; use Millions) or custom format strings
Sort	`…` menu → Sort axis (by value or by another field)
Colour a highlighted series	Format → Colors → per-category colours or conditional
Titles that state the insight	General → Title → fx (dynamic title from a measure)
Accessibility	General → Alt text; keep contrast high; do not rely on colour alone
Copy formatting	Home → **Format painter**
Align / distribute	Format → Arrange → Align and distribute; hold `Shift` to multi-select
Group visuals	Select several → Format → Group

TIP **Dynamic titles**

Create a measure `Title Sales = "Sales by " & SELECTEDVALUE ( dim_territory[country], "all countries" ) & " in " & SELECTEDVALUE ( dim_date[year], "all years" )` and bind it in Title → fx → *Field value*. The title now reflects the user's selections.

Filters: three levels



The three filter levels. Visual filters affect one visual, page filters one page, report filters every page.

Use the **Filters pane** (right side). Filter types:

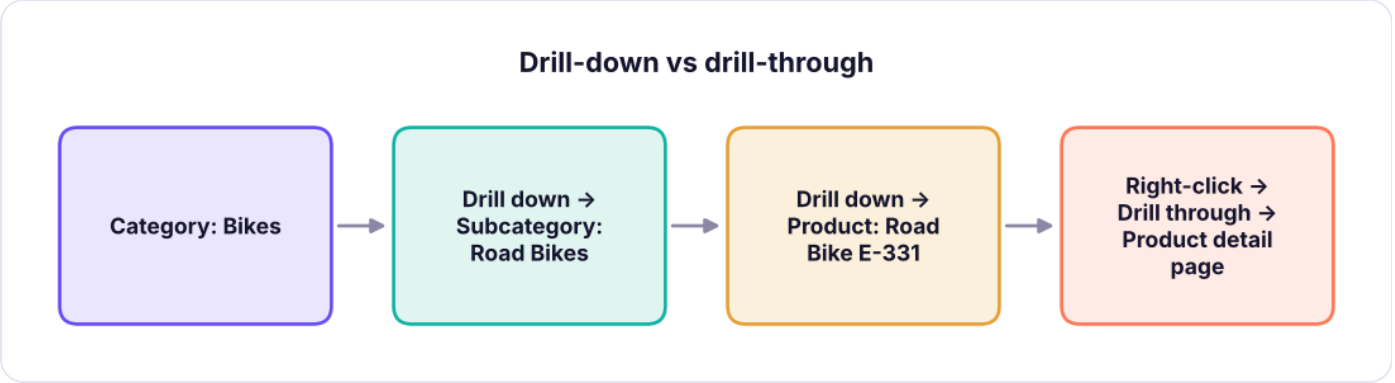
Filter type	Behaviour
Basic	Tick values from a list
Advanced	"contains", "starts with", "is greater than", with AND/OR
Top N	Top/Bottom N items by a measure (even one not in the visual)
Relative date	"Last 6 months", "this year", "next 7 days"
Relative time	Last 4 hours etc.

Tips: **Require single selection** in a slicer; **Sync slicers** across pages (View → Sync slicers); the **"Include" / "Exclude"** options on a selection filter in or out; lock the filters pane or hide it for consumers.

Edit interactions

By default clicking a bar **cross-filters** other visuals. Select the source visual → Format → **Edit interactions** → then on each target visual click the **Filter**, **Highlight** or **None** icon. Example: a KPI card for "overall sales" should ignore the clicks on the country chart (set to None), while the product chart should filter.

Drill-down and drill-through



Drill-down moves through a hierarchy inside one visual. Drill-through jumps to another page that is filtered by the selected item.

- **Drill-down**: needs a hierarchy (Lesson 2). Use the arrow buttons at the top of the visual: ↑ drill up, ↓ drill mode, ⇊ go to the next level in the hierarchy, ⇈ expand all.
- **Drill-through**: create a **detail page** → in the Visualizations pane, drag `product` into **Drill through → Add drill-through fields here** → Power BI adds a back button automatically. Users right-click a product in any visual → **Drill through → Product detail**. Use it for "customer 360", "order detail" and "product detail" pages. **Keep all filters** keeps slicer context.
- **Cross-report drill-through** links to a separate report in the Service.

Report-page tooltips

A **tooltip page** shows a mini-report when hovering over a data point.

STEP BY STEP **Build one**

1. Add a new page; Format page → Page information → **Allow use as tooltip** on; Page size → **Tooltip**.
2. Add visuals (a KPI, a small trend line, top 3 products) using fields that respond to the hovered item.
3. In the main visual: Format → General → **Tooltips** → Type **Report page** → choose the page.
4. Hide the tooltip page from navigation (right-click the tab → Hide page).

Buttons, bookmarks and navigation

Feature	What it does	Use
Buttons (Insert → Buttons)	Back, Bookmark, Page navigation, Q&A, Web URL, Apply all slicers, Clear all slicers	Menus, navigation bars, reset filter
Bookmarks (View → Bookmarks)	Save the current state (filters, slicer selections, visibility, spotlight) of a page	Story-telling views, toggle between visuals
Selection pane (View → Selection)	List, name, show or hide every object on a page	Required for bookmark toggles
Page navigator (Insert → Navigator)	Auto-generated tab buttons for pages or bookmarks	Menu
Conditional visibility	Bookmarks that show/hide visuals	"Chart ⇄ Table" toggle

STEP BY STEP

Chart / table toggle with bookmarks

1. Place a column chart and a table on top of each other.
2. Selection pane: hide the table → View → Bookmarks → **Add** → name it "Chart view"; make the table visible and hide the chart → Add → "Table view". Untick **Data** in each bookmark's options so slicers are not reset.
3. Insert two buttons and set Action → **Bookmark** → choose the matching bookmark.

Conditional formatting

Where	How
Table / matrix cells	Format → Cell elements → Background colour / Font colour / **Data bars** / **Icons** / Web URL → fx → *Gradient*, *Rules* or *Field value*
Column / bar colour	Format → Colors → fx → rules (green above target, red below)
Card text colour or title	fx on the callout/title
Based on a measure	Create a measure returning a hex colour: `Margin Colour = IF ( [Profit Margin %] ⩾ 0.35, "#1E8E5A", "#C0392B" )` then Field value

Subcategory	Sales (₹ M)	Profit (₹ M)	Margin %
Mountain Bikes	161.9	54.9	33.9%
Road Bikes	155.7	56.1	36.0%
Touring Bikes	149.2	55.2	37.0%
Frames	76.1	27.3	35.8%
Wheels	33.4	10.5	31.3%
Helmets	8.2	2.7	33.1%
Jerseys	5.8	2.2	37.2%
Gloves	2.7	0.8	30.8%
Bottles	1.6	0.5	34.0%

A matrix with conditional formatting: a heat-map background on sales and traffic-light colours on margin (rules: ≥ 36% green, otherwise amber/red).

Themes

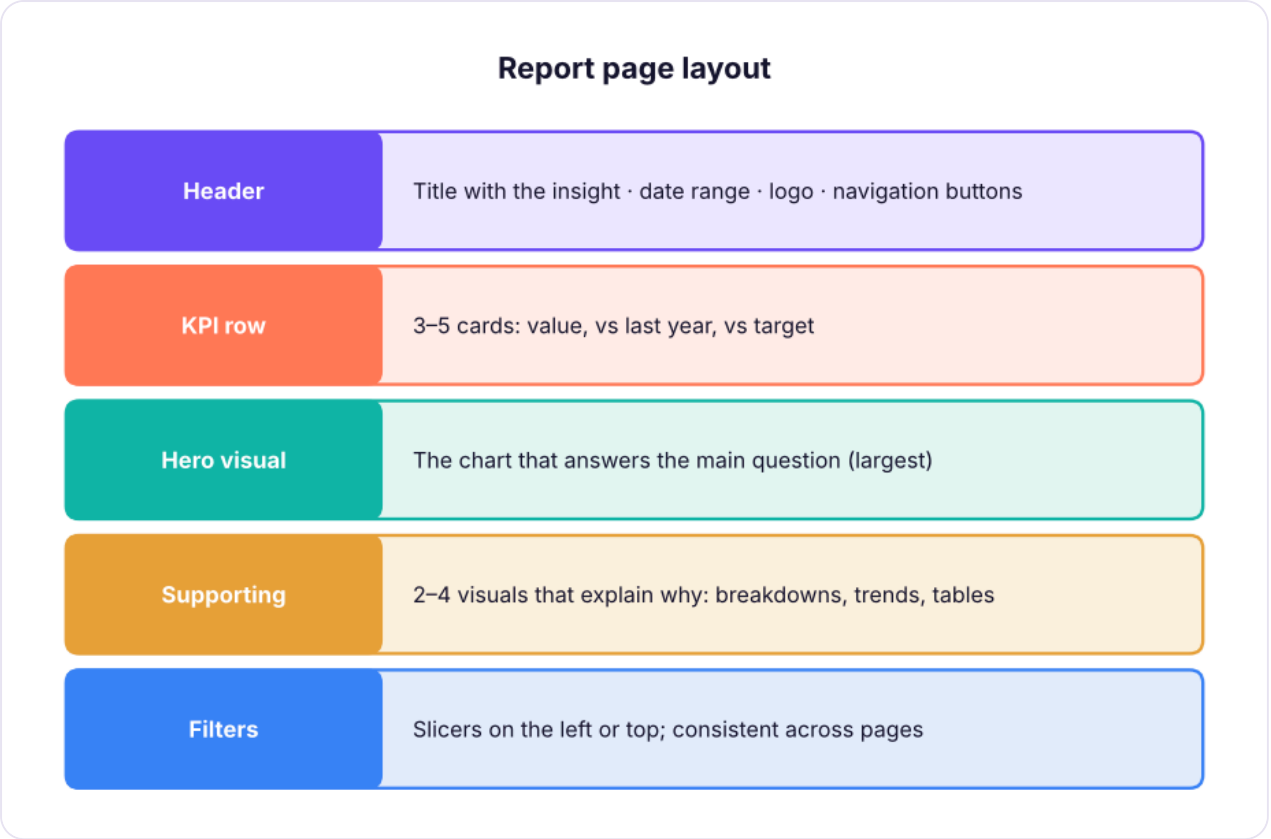
View → Themes offers built-in themes; you can download community themes or **Customize current theme** (colours, fonts, defaults). A **theme JSON file** (View → Themes → Browse for themes) applies one consistent look to every visual:

JSON

```
{
  "name": "SkillNavora",
  "dataColors": ["#6C4CF5", "#FF7A59", "#12B5A6", "#E9A23B", "#3B82F6", "#E04F8A"],
  "background": "#FFFFFF", "foreground": "#17142E", "tableAccent": "#6C4CF5",
  "textClasses": { "title": { "fontFace": "Segoe UI Semibold", "fontSize": 14, "color":
"#17142E" } }
}
```

Apply the company theme to every report so they look consistent, and edit colours in one place.

Report design principles



A layout that works: headline KPIs on top, the main insight chart large, supporting visuals below, filters on one side, consistent spacing.

1. **Start from the audience and decision**, not from the data: what will they do differently?
2. **Hierarchy**: most important top-left; big numbers first; details lower.
3. **Less is more**: 5–8 visuals per page; remove gridlines, borders and chart junk.
4. **Consistent colour logic**: one brand colour plus one highlight; red and green only for bad/good; the same colour always means the same thing.
5. **Align and space** visuals on a grid; same sizes for similar charts.
6. **Label for meaning**: insight titles, units (₹, %), clear axis titles, "as of" dates.
7. **Accessible**: contrast, alt text, tab order (Format → Tab order), do not use colour alone, test on mobile layout (View → Mobile layout).
8. **Performance**: fewer visuals, summarised data, no huge tables; check Optimize → Performance analyzer.

Q&A, images and mobile

- **Q&A visual**: users type "total sales by country in 2025" and get a chart. Improve it with **synonyms** (Model view → Linguistic schema / Q&A setup).

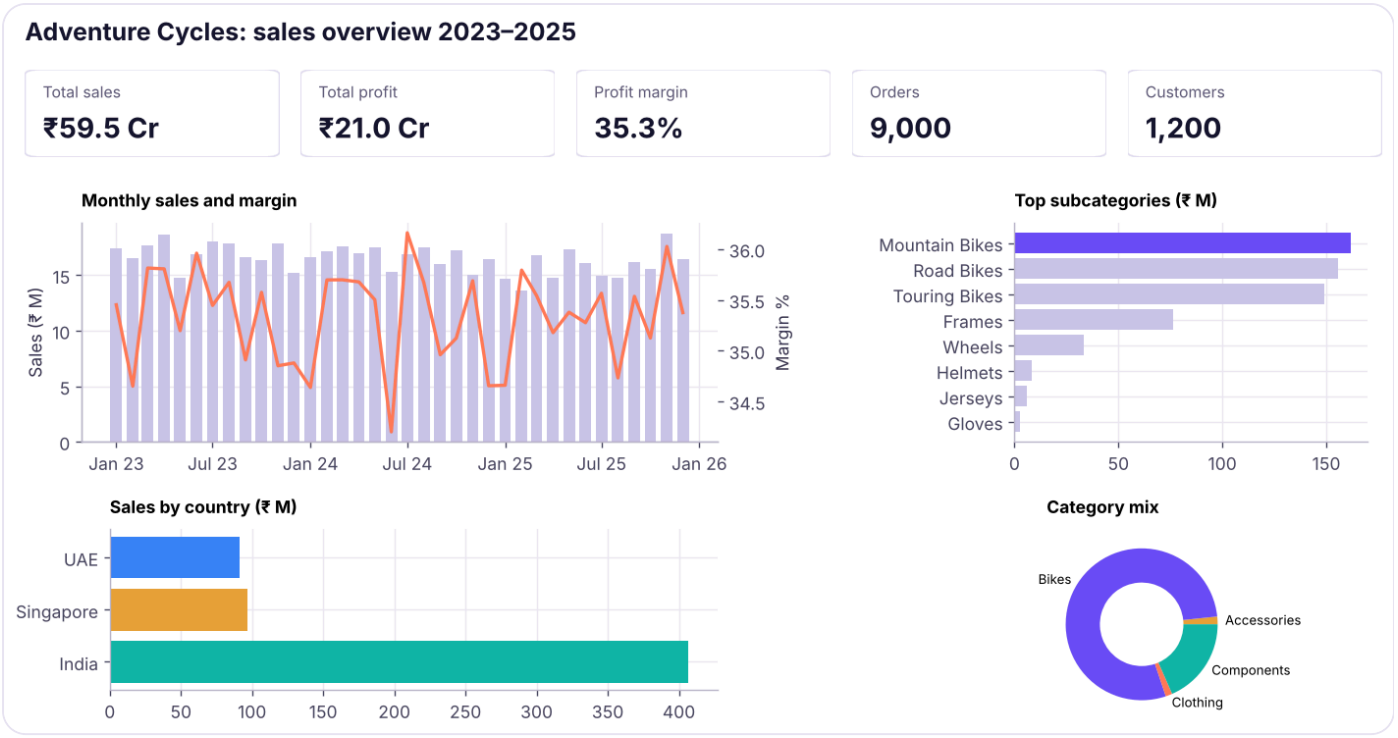
- **Images and logos**: Insert → Image (or text box). For images **in data** (product photos), set the column's **Data category → Image URL**.
- **Mobile layout**: View → Mobile layout; drag key visuals into a phone-shaped canvas.
- **Accessibility and exporting**: export to PDF or PowerPoint (File → Export), analyse in Excel, subscribe in the Service.

> **2026 UPDATE** **Copilot and AI visuals in 2026**
>
> Copilot can generate a report page from a prompt, suggest visuals for your fields, write summaries ("Summarise this page") and answer questions about the data. **Key influencers**, **Decomposition tree** and **Smart narrative** are built-in AI visuals; **anomaly detection** (Analytics pane) flags unusual spikes. Always verify AI-generated visuals and text against real numbers.

A mini case: the Adventure Cycles overview page

Plan the page before you build it:

Element	Fields / measure	Visual
Title	Dynamic title measure	Text box
KPI row	`Total Sales`, `Total Profit`, `Profit Margin %`, `Orders`, each with vs-last-year	4–5 cards
Hero	Monthly sales and profit margin	Line + column
Breakdown 1	Sales by category (Top N subcategories)	Bar
Breakdown 2	Sales by country	Map or bar
Drill-through	Product detail page	Table + KPIs
Filters	Year, Country, Category	Slicers



The Adventure Cycles overview page from the real data: KPI cards, monthly trend, subcategory ranking and country share.

Summary

KEY TAKEAWAYS

- Choose visuals from the question; format with insight titles, units, sorting and accessible colours.
- Filters work at **visual, page and report** level (Basic, Advanced, Top N, Relative date); use slicers, sync them across pages, and **Edit interactions** to control cross-filtering.
- **Drill-down** needs a hierarchy; **drill-through** jumps to a detail page; **tooltip pages** show mini-reports on hover.
- **Buttons + bookmarks + the Selection pane** build navigation and chart/table toggles.
- Use **conditional formatting** (rules, gradients, field value from a measure) and **themes** for consistency.
- Design for the decision: hierarchy, few visuals, consistent colour logic, alignment, labels, accessibility and mobile layout.

Quick quiz

6 questions · tap an answer to check it

1. Which filter level affects every page of a report?

A Visual-level

B Page-level

C Report-level (All pages)

D Slicer-level

Answer: C. Report filters apply to all pages.

2. How do you jump from a product in a chart to a detail page for that product?

A Drill-down

B Drill-through (create a drill-through page with the product field)

C Add a slicer

D Use a tooltip

Answer: B. Drill-through passes the selected value as a filter to another page.

3. What do bookmarks capture?

A Only the page name

B The state of a page: filters, slicers, visibility and more

C The data model

D The theme

Answer: B. Bookmarks save a view that buttons can switch to.

4. Which feature stops a card from reacting when you click a bar chart?

A Sync slicers

B Edit interactions → set the card to None

C Drill-through

D Alt text

Answer: B. Edit interactions controls how visuals filter each other.

5. When is a donut chart appropriate?

A With 12 categories

B With 2–3 parts of a whole

C To show a time trend

D To compare 20 products

Answer: B. More than a few slices are hard to compare; use bars.

6. What is the benefit of a theme JSON file?

A One consistent set of colours and fonts applied to all visuals and reusable across reports

B It refreshes data

C It creates relationships

D It adds measures

Answer: A. Themes standardise the look and make branding quick.

EXERCISE

Build the overview page

Using the Adventure Cycles model: (1) build the overview page above; (2) add a **Year** slicer and a **Country** slicer; (3) create a drill-through page for **Product** with sales, profit, margin and a monthly trend; (4) add a bookmarked toggle between "Sales" and "Profit" using a field parameter or bookmarks; (5) apply conditional formatting to the margin in a matrix and write a dynamic title.

Show solution (worked answer)

(1) Cards for the five measures; line + clustered column for monthly sales and margin; bar for top subcategories; bar for country; donut for category mix. (2) Slicer visuals with `dim_date[year]` (tile style) and `dim_territory[country]` (dropdown). (3) New page → drag `dim_product[product]` into Drill through; add a back button. (4) Modeling → New parameter → Fields → add `Total Sales`, `Total Profit`; use the parameter on the chart values; place its slicer as a toggle. (5) Matrix of `subcategory` × `country` with `Profit Margin %`; Cell elements → Background colour → Rules; title measure with `SELECTEDVALUE`.

INTERVIEW CORNER

Questions you may be asked

Q: How do you design an effective Power BI dashboard? Start from the audience and decision; put headline KPIs top-left; use few visuals with insight titles; consistent colour and spacing; allow drill-down/through; test with users; check performance and accessibility.

Q: Drill-down vs drill-through? Drill-down navigates levels of a hierarchy within a visual; drill-through navigates to another page filtered by the selected item.

Q: What are the filter levels and how do slicers differ? Visual, page and report filters are set by the author in the Filters pane; slicers are on-canvas controls that users change, and they can be synced across pages.

Q: How would you let users toggle between a chart and a table? Two bookmarks that show/hide the visuals through the Selection pane, and two buttons linked to those bookmarks.

POWER BI · LESSON 6

Power BI Service, Connectivity Modes, Security & Capstone

Take a report from Desktop to the people who need it. Understand Import, DirectQuery, Live connection, composite models and Direct Lake, publish to the Service, schedule refresh with a gateway, share through workspaces and apps, secure data with row-level security, tune performance, work with Copilot and Fabric, and complete the Adventure Cycles capstone.

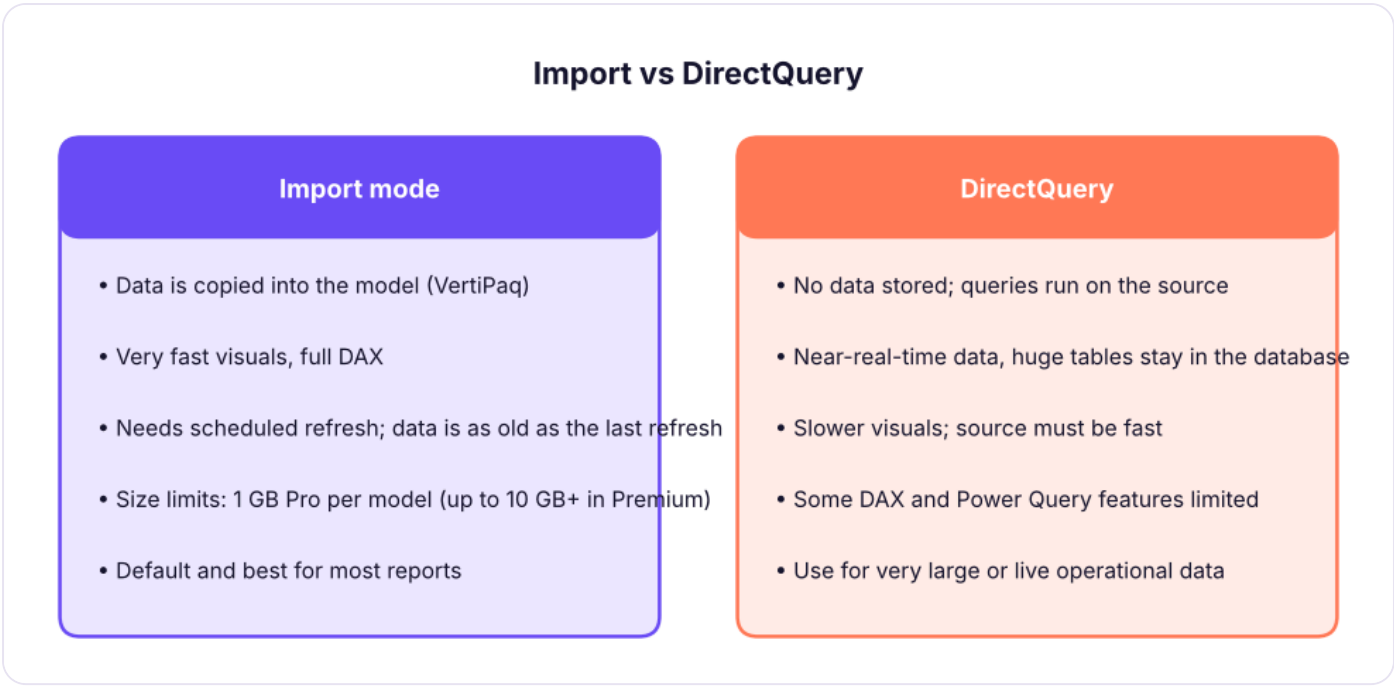
Power BI class 4 | ✦ Power BI Service & Microsoft Fabric 2026 | Advanced | ⏱ 150 min

By the end of this lesson you can

- Compare Import, DirectQuery, Live connection, Composite and Direct Lake modes and choose one
- Publish a report, configure scheduled refresh and a gateway, and share via workspaces, apps and links
- Implement and test row-level security (static and dynamic) with DAX filters
- Use Performance Analyzer and best practices to speed up a model
- Describe Microsoft Fabric, Copilot and governance basics, and deliver an end-to-end capstone

Connectivity modes

How does Power BI get its data: **copy it** into the model (Import) or **ask the source each time** (DirectQuery)?



Import copies data into Power BI's compressed in-memory model; DirectQuery sends a query to the source for every interaction.

Mode	Data stored in Power BI?	Refresh	Typical use
Import	Yes (compressed copy)	Scheduled (up to 8/day Pro, 48/day Premium)	Most reports; Excel, CSV, SQL
DirectQuery	No	Live per interaction	Large SQL databases, near-real-time dashboards
Live connection	No: connects to an existing **semantic model** (Analysis Services or a published Power BI dataset)	Live	One governed model shared by many reports
Composite model	Mix: some tables Import, some DirectQuery, plus other semantic models	Both	Large fact in DirectQuery, small dimensions imported (dual storage mode)
Direct Lake (Fabric)	Reads Delta tables in OneLake directly	Automatic on data change	Import-like speed with no copy; Fabric only

TIP **A practical rule**

Start with **Import**. Move to DirectQuery or Direct Lake only if data volume or freshness requirements force you to. Reduce the data first: only needed columns, aggregated tables, filtered date range.

Publishing and the Service

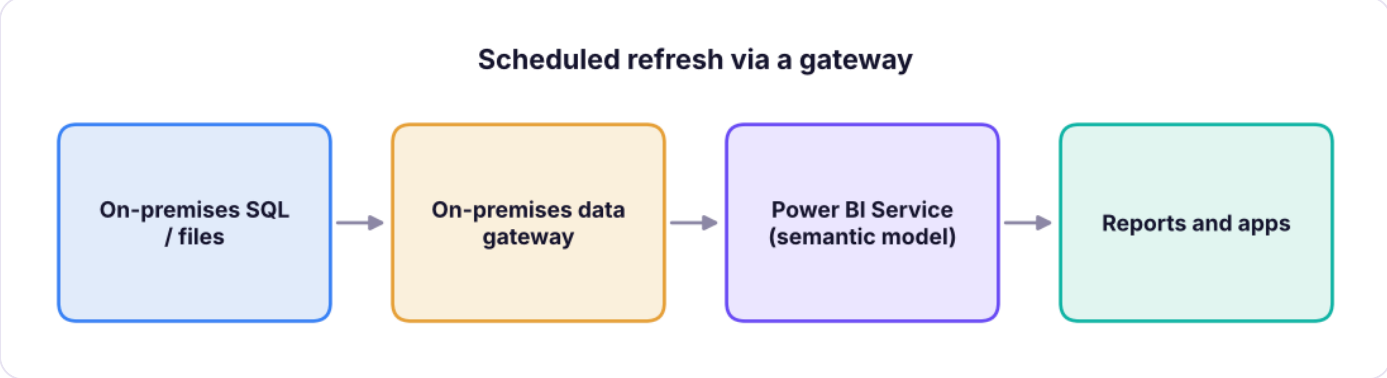
STEP BY STEP **Publish a report**

1. In Desktop: **Home → Publish** → sign in with a work/school account → choose a **workspace** (not "My workspace" if you want to share).
2. The Service now has a **semantic model** (the data and model) and a **report** (the visuals): the same objects in two items.
3. Open the report in the browser, check it, and **Share** with people or add it to an **app**.
4. Configure **scheduled refresh** (Settings → Semantic model → Scheduled refresh) with credentials; for on-premises sources install a **gateway**.

Workspaces, apps and sharing

Concept	Meaning
Workspace	A shared container for semantic models, reports, dashboards. Roles: **Admin, Member, Contributor, Viewer**
App	A packaged, read-only, curated set of reports/dashboards published to a wider audience (best way to distribute)
Share link	Give a person or group access to one report (they need appropriate licence)
Dashboard	One page of **pinned tiles** from several reports; supports alerts and mobile
Subscriptions & alerts	Email snapshots on a schedule; data alerts on card/gauge thresholds
Export / Analyze in Excel / Embed	PDF/PowerPoint export, live pivot tables on the model, secure embedding in portals
Deployment pipelines	Move content Dev → Test → Prod with controlled changes
Endorsement, sensitivity labels, lineage view	Governance: certified datasets, data classification, impact analysis

Refresh and gateways



A scheduled refresh for on-premises data goes through a gateway installed inside the company network.

- **Cloud sources** (Azure SQL, SharePoint Online, OneDrive files) refresh directly without a gateway.
- A **standard gateway** serves many users; a **personal gateway** is for one person's testing.
- **Incremental refresh** (right-click a table → Incremental refresh) loads only new or changed partitions (for example the last 7 days) instead of the whole fact table: essential for big tables. It needs `RangeStart` and `RangeEnd` parameters (Power Query).

- Refresh failures (expired credentials, changed schema, gateway offline) appear in the refresh history; enable failure notifications.

Row-level security (RLS)

RLS restricts the rows a user can see: a country manager sees only their country.

Static RLS

STEP BY STEP **Create roles in Desktop**

1. **Modeling → Manage roles → Create**. Name the role `India`.
2. Select the table `dim_territory` and add a DAX filter: `[country] = "India"`.
3. Create roles `UAE` and `Singapore` the same way. **Save**.
4. **View as** (Modeling → View as) → choose a role to test what that person sees.
5. Publish; in the Service: **Semantic model → Security** → add users or groups to each role.

Because filters flow from `dim_territory` to `fact_sales` through the relationship, the India role sees only India's sales everywhere:

Python

```
import pandas as pd
D = "adventure_cycles/"
f = pd.read_csv(D + "fact_sales.csv"); t = pd.read_csv(D + "dim_territory.csv"); m =
f.merge(t, on="territory_key")
role_view = m.groupby("country").agg(orders=("order_number", "count"), sales=
("sales_amount", "sum"))
role_view["sales"] = (role_view.sales / 1e6).round(1)
role_view.rename(columns={"sales": "sales (₹ M) visible to that role"}).assign(**{"all-
company total (₹ M)": round(f.sales_amount.sum() / 1e6, 1)})
```

OUTPUT

	orders	sales (₹ M) visible to that role	all-company total (₹ M)
country			
India	6135	406.9	594.6
Singapore	1487	96.7	594.6
UAE	1378	91.0	594.6

3 rows × 3 columns

Dynamic RLS

One role for everyone, driven by a **security table** mapping user emails to territories:

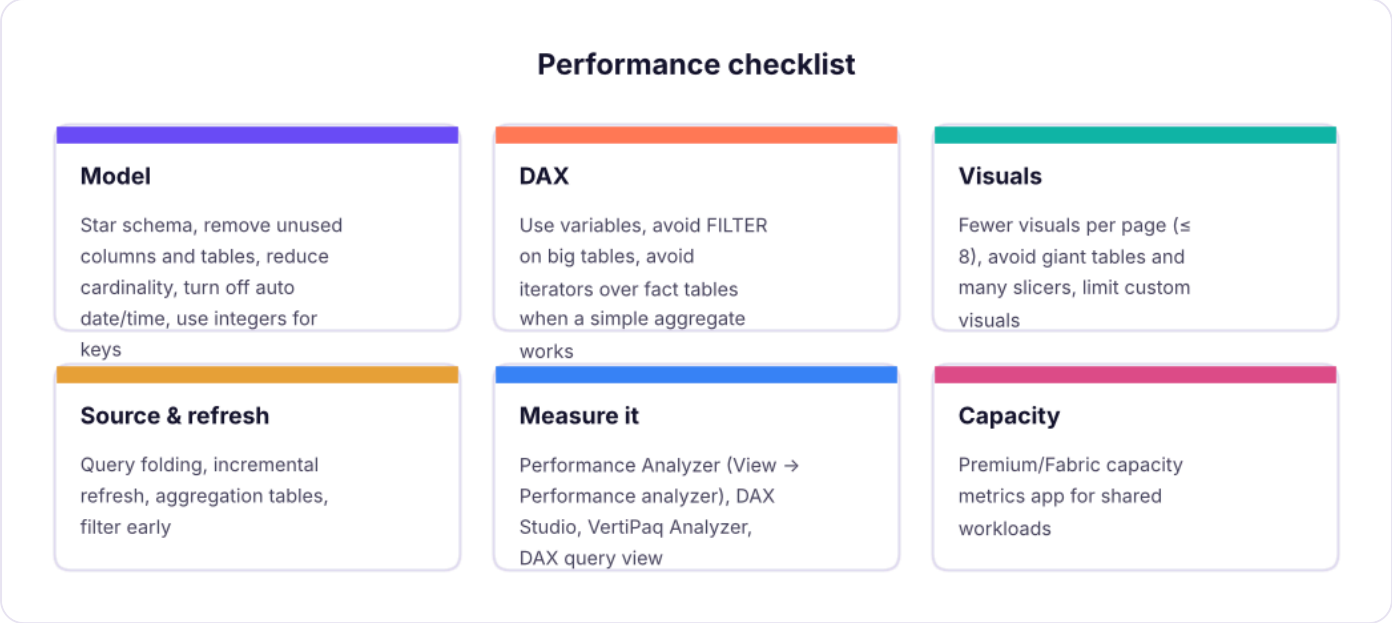
```dax
-- Table 'UserSecurity'(user_email, territory_key); role filter on dim_territory:
[territory_key]
    IN CALCULATETABLE ( VALUES ( UserSecurity[territory_key] ),
                        UserSecurity[user_email] = USERPRINCIPALNAME () )
```

`USERPRINCIPALNAME()` returns the signed-in user's email. Add a person to the table and they get access automatically: no role edits.

WATCH OUT **RLS pitfalls**

- RLS applies to **Viewers** only. Workspace Admins, Members and Contributors can see all data.
- RLS does not hide column names or the existence of tables; use **object-level security** for that.
- Filters must flow along relationships: apply RLS to a **dimension**, not the fact, where possible.
- RLS with DirectQuery depends on the source's behaviour; test with *View as*.
- Do not store secrets or personal data you do not need in the model. In India, handling personal data is governed by the DPDP Act 2023.

Performance tuning



Where Power BI report time goes, and where to look.

Performance Analyzer (View → Performance analyzer → Start recording → refresh visuals) shows, for each visual, the time spent in **DAX query**, **Visual display** and **Other**. If DAX dominates, optimise the measure or model; if display dominates, simplify the visual.

Practical wins:

Problem	Fix
File is huge	Remove columns you do not use; store IDs as integers; split date and time; use `Whole number`/`Fixed decimal` rather than text
Slow card or matrix	Rewrite measures with variables; avoid nested `FILTER ( ALL ( bigTable ) )`
Slow refresh	Query folding, incremental refresh, filter at source
Slow DirectQuery	Aggregations table (Import) in front of the big table; index the source
Many identical measures	Calculation groups or one base measure referenced by others

Microsoft Fabric, Copilot and AI

Microsoft Fabric unifies data engineering, data factory, data warehouse, data science, real-time analytics and Power BI over one data lake called **OneLake** (Delta/Parquet files). Power BI's semantic models can use **Direct Lake** over lakehouse tables. For data professionals: learn SQL (this track), Python (next track), and then Fabric notebooks, pipelines and lakehouses.

Fabric item	Role
Lakehouse / Warehouse	Stores data (files + SQL endpoint)
Dataflow Gen2	Power Query in the cloud
Notebook	Python/Spark for data science and engineering
Pipeline	Orchestrates data movement
Semantic model	The Power BI model (formerly "dataset")
Eventstream / Real-time	Streaming data

Copilot in Power BI (needs a Fabric or Premium capacity): *create a report page* from a prompt, *suggest measures* and *write DAX*, *summarise* a page or report, and *answer questions* about the data. **Prep data for AI** (add descriptions, synonyms, verified answers) improves the answers. Treat it as a fast assistant, not an authority.

Governance and good practice

1. **One certified semantic model per subject area**, many thin reports on top (live connection).
2. Name items consistently (`Sales - Overview`, `Sales_Model`); keep an **owner** and a description.
3. Version control: **PBIP** + Git, deployment pipelines, a test workspace.
4. Document measures (description property) and keep a measure dictionary.
5. Limit **who can export** data; use sensitivity labels.
6. Monitor usage (usage metrics report) and remove unused reports.
7. Train users: the best dashboard is useless if nobody trusts or understands it.

Capstone: Adventure Cycles performance review

CASE STUDY

The brief

The sales director wants a 3-page Power BI report for the Adventure Cycles leadership team and country managers: **(1) Executive summary**, **(2) Products and customers**, **(3) Country detail (drill-through)**. It must show YTD and vs-last-year numbers, let managers filter by year, country and category, and each country manager must see **only their own country**.

Deliverables checklist

Step	What to produce
1. Requirements	Audience, questions, KPIs, refresh frequency, security
2. Data	Load the five CSV tables; clean in Power Query
3. Model	Star schema, relationships, marked date table, hierarchies (Lessons 1–2)
4. Measures	Sales, Profit, Margin %, Orders, AOV, YTD, LY, YoY %, Rank, Top 5 + Others (Lessons 3–4)
5. Report	Three pages, theme, tooltips, drill-through, bookmarks, mobile layout (Lesson 5)
6. Security	RLS roles by country, tested with *View as*
7. Publish	Workspace, scheduled refresh, app, subscriptions (this lesson)
8. Documentation	One-page "how to read this report" and a measure dictionary

The analysis the report must support: verify your page numbers against this independent calculation.

Python

```
m = f.merge(t, on="territory_key").merge(pd.read_csv(D + "dim_product.csv"),
on="product_key"); m["profit"] = m.sales_amount - m.total_cost
m["order_date"] = pd.to_datetime(m.order_date); m["year"] = m.order_date.dt.year
kpi = m.groupby("year").agg(sales=("sales_amount", "sum"), profit=("profit", "sum"),
orders=("order_number", "nunique"))
kpi["margin_%"] = (kpi.profit / kpi.sales * 100).round(1); kpi["YoY_sales_%"] =
(kpi.sales.pct_change() * 100).round(1)
kpi[["sales", "profit"]] = (kpi[["sales", "profit"]] / 1e6).round(1)
kpi
```

OUTPUT

	sales	profit	orders	margin_%	YoY_sales_%
year					
2023	204.2	72.2	3004	35.4	NaN
2024	200.4	70.8	3022	35.3	-1.9
2025	189.9	67.2	2974	35.4	-5.2

3 rows × 5 columns

Python

```
c = m.groupby("country").agg(sales=("sales_amount", "sum"), profit=("profit", "sum"))
c["margin_%"] = (c.profit / c.sales * 100).round(1); c["share_%"] = (c.sales /
c.sales.sum() * 100).round(1); c[["sales", "profit"]] = (c[["sales", "profit"]] /
1e6).round(1)
bike_share = (m[m.category == "Bikes"].sales_amount.sum() / m.sales_amount.sum() *
100).round(1)
print(c.to_string()); print(f"\nBikes are {bike_share}% of total sales; Components
{m[m.category == 'Components'].sales_amount.sum() / m.sales_amount.sum() * 100:.1f}%.")
```

OUTPUT

```
           sales  profit  margin_%  share_%
country
India      406.9   144.1      35.4     68.4
Singapore   96.7    34.1      35.2     16.3
UAE         91.0    32.0      35.1     15.3

Bikes are 78.5% of total sales; Components 18.4%.
```

KEY TAKEAWAYS **What a good leadership summary says**

1. **Sales are shrinking, not growing**: ₹204.2 M (2023) → ₹200.4 M (2024, −1.9%) → ₹189.9 M (2025, −5.2%), while the margin holds steady at about 35%. The problem is **volume or demand**, not pricing or cost.
2. **Bikes** generate 78.5% of sales and **India** 68.4%, so any decline in Indian bike sales moves the whole company: a concentration risk.
3. Singapore (16.3% of sales) and the UAE (15.3%) earn the same ~35% margin as India, so growth there is as profitable as at home.
4. Recommended next steps: find where the 2025 decline comes from (by category, country and month, using the YoY measures from Lesson 4), set recovery targets per country, and track them monthly in the report.

(Verify each statement against the tables above before you present it; the numbers, not the template, are the story.)

Summary

KEY TAKEAWAYS

- **Import** is fast and the default; **DirectQuery** keeps data in the source (slower, near-live); **Live connection** reuses a shared semantic model; **Composite** mixes modes; **Direct Lake** reads Fabric Delta tables directly.
- Publish to a **workspace**, schedule refresh (gateway for on-premises, incremental refresh for big facts), distribute through **apps**, and use subscriptions and alerts.
- **RLS** filters rows by role (static) or by the signed-in user (dynamic, `USERPRINCIPALNAME()`); it applies to Viewers and must be tested with *View as*.
- Tune performance with a star schema, lean columns, efficient DAX and **Performance Analyzer**.
- Fabric unifies the data stack; Copilot speeds up building; governance (certified models, pipelines, labels) keeps it trustworthy.

Quick quiz

6 questions · tap an answer to check it

1. Which mode stores a compressed copy of the data inside the Power BI model?

A Import

B DirectQuery

C Live connection

D Streaming

Answer: A. Import loads the data into the in-memory VertiPaq model.

2. When is an on-premises data gateway needed?

A For any Excel file

B To schedule refresh from data sources inside a company network

C To publish reports

D To use slicers

Answer: B. Gateways let the cloud service reach on-premises sources.

3. A user with the Admin role in the workspace is also in an RLS role. What do they see?

A Only the role's rows

B All data, because RLS does not apply to Admin, Member and Contributor roles

C Nothing

D Only their own data

Answer: B. RLS applies to Viewers; test with View as.

4. What does incremental refresh do?

A Refreshes slicers only

B Refreshes only new or changed partitions of a large table

C Deletes old data

D Converts the model to DirectQuery

Answer: B. It speeds up refresh by loading only recent data.

5. Which tool shows time spent in DAX vs visual display?

A Model view

B Performance Analyzer

C Selection pane

D Bookmarks

Answer: B. Performance Analyzer breaks down each visual's load time.

6. Which function enables dynamic row-level security by user?

A `TODAY ()`

B `SELECTEDVALUE ()`

C `USERPRINCIPALNAME ()`

D `RELATED ()`

Answer: C. It returns the logged-in user's email for comparison with a security table.

EXERCISE

Deploy and secure

(1) Create RLS roles for the three countries and test each with *View as*. (2) Create a dynamic RLS design using a `UserSecurity` table (write the table and the DAX filter). (3) List five changes you would make to a 2 GB model that refreshes in 45 minutes. (4) Write the checklist you would give a colleague before publishing a report to leadership.

Show solution (worked answer)

(1) Manage roles → filter `dim_territory[country]`; View as each role and confirm the totals equal the pandas numbers above. (2) Table: `user_email, territory_key`; filter on `dim_territory`: `[territory_key] IN CALCULATETABLE ( VALUES ( UserSecurity[territory_key] ), UserSecurity[user_email] = USERPRINCIPALNAME () )`. (3) Remove unused columns; star schema and integer keys; incremental refresh on the fact; push transformations to the source (query folding); aggregate to the needed grain; disable auto date/time. (4) Numbers reconciled with source; filters and slicers tested; titles state insights; RLS tested; refresh scheduled and tested; owner and description added; performance checked; accessibility and mobile layout checked; app published with notes.

INTERVIEW CORNER

Questions you may be asked

Q: Import vs DirectQuery? Import copies and compresses data into the model (fast, needs refresh, size limits); DirectQuery queries the source live (fresh, handles huge data, slower, DAX limitations).

Q: What is row-level security and how do you implement it? Restricting the rows a user can see: define roles with DAX filters on tables (static), or use USERPRINCIPALNAME and a security table (dynamic); assign users in the Service; test with View as.

Q: How do you improve a slow Power BI report? Measure with Performance Analyzer; fix the model (star schema, fewer columns), optimise DAX, simplify visuals, use aggregations/incremental refresh, and reduce data volume.

Q: What is a semantic model and why use live connections? The data model (tables, relationships, measures); multiple reports can connect live to one governed model so everyone uses the same definitions.

POWER BI · LESSON 7

Microsoft Fabric: Lakehouse, Warehouse, Dataflows and Semantic Models

Understand Microsoft Fabric as one platform. Load data into a lakehouse, shape it in a warehouse or dataflow, publish a semantic model that Power BI reports read from, and keep the whole chain governed and refreshed.

Gap-fill lesson | Microsoft Fabric 2026 | Advanced | 150 min

By the end of this lesson you can

- Describe the Fabric workspace items: lakehouse, warehouse, dataflow, notebook and semantic model
- Choose between a lakehouse and a warehouse for a given workload
- Load and shape data with a dataflow and with the SQL analytics endpoint
- Publish a semantic model and connect a Power BI report to it
- Explain OneLake, shortcuts and the refresh chain

What Fabric is

Microsoft Fabric is a single platform for data work, built on one storage layer called **OneLake**. Instead of separate tools for storage, pipelines, warehousing and reporting, every item in Fabric reads from the same copy of the data. Power BI is one of the experiences inside Fabric.

For a Power BI analyst, the important items are:

Item	What it is	Typical use
Lakehouse	Files and tables stored in OneLake, read with SQL or Spark	Large data, mixed files and tables
Warehouse	A SQL-first store for structured tables	Star schemas for reporting
Dataflow Gen2	No-code Power Query data preparation	Cleaning and loading without code
Notebook	Python, PySpark or SQL code	Advanced transformations and data science
Semantic model	The model (tables, relationships, DAX measures) that reports use	Shared, governed measures
Report	The Power BI visuals	What people look at

CASE STUDY **One sales model, many reports**

A retail company keeps sales data in OneLake. Finance, marketing and operations each want their own report, but they must all agree on what "revenue" means. The answer is one semantic model in a Fabric workspace that all three reports connect to, so the measure is written once and changed in one place.

OneLake and shortcuts

OneLake is the single logical data lake for the organisation. Each workspace has a lakehouse or warehouse, and the data is stored once. A **shortcut** points to data in another location (another lakehouse, or Azure storage, or Amazon S3) without copying it. Use shortcuts to avoid duplicating large tables across workspaces.

::::tip Shortcut, do not copy A copy drifts out of date and doubles storage. A shortcut always reads the current data. Copy only when you need a snapshot, for example for an audit. ::::

Lakehouse versus warehouse

Both store tables in OneLake. The choice depends on the workload:

- **Lakehouse**: keeps files (CSV, JSON, Parquet) alongside tables. Spark notebooks and SQL both work on it. Good for data engineering and mixed data.
- **Warehouse**: SQL-only tables with full T-SQL support, including stored procedures and fine-grained permissions. Good for structured reporting and teams that think in SQL.

A lakehouse also exposes a **SQL analytics endpoint**, which lets you query its tables with SQL. Many teams therefore land raw data in a lakehouse, then build reporting tables in a warehouse or view them through the endpoint.

Loading data with a dataflow

A dataflow uses Power Query, the same tool as Power BI Desktop, to load and shape data in the service. It is the right choice when the transformation is ordinary cleaning and the team does not write code.

Steps:

1. In the workspace, select **New item > Dataflow Gen2**.
2. Add a source, for example a CSV file uploaded to the lakehouse.
3. Apply transformations: change types, remove blank rows, merge with a customer table.
4. Set the output destination to a lakehouse or warehouse table.
5. Publish, then run the dataflow once to check the output.

The output is a table in OneLake. Any report can read it, which is why dataflows suit shared, reusable cleaned data.

::::warn Keep logic in one place If the same cleaning happens in a dataflow, in a notebook and again in Power BI Desktop, the three versions will disagree. Choose one place for each transformation and document it. ::::

Building a star schema in the warehouse

Reporting works best on a star schema: one fact table of sales and several dimension tables of customers and products. Create the tables in the warehouse with a short script, then load them.

SQL

```sql
-- Dimension: customers (one row per customer)
CREATE TABLE dbo.dim_customer (
    customer_key   INT         NOT NULL,
    customer_name  VARCHAR(100),
    segment        VARCHAR(50),
    territory      VARCHAR(50)
);

-- Fact: sales (one row per order line, keys point to dimensions)
CREATE TABLE dbo.fact_sales (
    order_date     DATE,
    customer_key   INT,
    product_key    INT,
    quantity       INT,
    sales_amount   DECIMAL(12, 2),
    cost_amount    DECIMAL(12, 2)
);
```

The fact table holds numbers and keys. Descriptive text lives in dimensions. This shape keeps the semantic model small and fast, and it matches the star schema taught in the modelling lesson.

Publishing a semantic model

A semantic model is the Power BI dataset in Fabric. You create it from the warehouse or lakehouse tables, add relationships and measures, and publish it to the workspace.

1. Open the warehouse and choose **New semantic model**.
2. Select the tables to include, for example `fact_sales`, `dim_customer` and `dim_product`.
3. Confirm the relationships. They should run from each dimension (one side) to the fact table (many side).
4. Write the DAX measures in the model, not in individual reports, so every report uses the same definition:

DAX

```dax
Total Sales = SUM ( fact_sales[sales_amount] )
Total Cost  = SUM ( fact_sales[cost_amount] )
Gross Margin % = DIVIDE ( [Total Sales] - [Total Cost], [Total Sales] )
```

5. Save the model. Reports now connect to it in Power BI and see the same measures.

Connecting a report

From Power BI Desktop, choose **Get data > Power BI semantic models** and select the published model. The report is built on top of the shared model: visuals and measures come from the model, while layout and filters belong to the report.

This separation is the key idea. A change to the model's measure updates every report that uses it, without editing each one. The reverse also matters: a report should not define its own revenue measure, or the numbers will drift.

Refresh and the chain of dependencies

Data flows through a chain: source, then dataflow or notebook, then lakehouse or warehouse, then semantic model, then report. A refresh must run in order. If the dataflow fails, the model shows yesterday's data, and the report shows yesterday's numbers without any visible error.

Practical rules:

- Schedule the upstream item first, and the semantic model refresh after it, with a short gap.
- Check the refresh history after each run, not only when a report looks wrong.
- Use a notification on failure so that the team hears about it first.

Governance basics in Fabric

- **Workspace roles** (Admin, Member, Contributor, Viewer) control who can change items. Give analysts Contributor or Member only where they need to build.
- **Endorsement** marks a semantic model as Promoted or Certified, so people know which model to trust.
- **Lineage view** shows how a report depends on the model, the warehouse and the source, which speeds up impact checks before a change.
- **Sensitivity labels** can classify data, for example as Confidential, and follow it through the chain.

::::key What to remember - Fabric keeps all data in one place, OneLake, and every item reads from it. - Use shortcuts instead of copies, and lakehouses for files and mixed data, warehouses for structured reporting. - Dataflows suit no-code cleaning; notebooks suit code; both write tables to OneLake. - Define measures once in the semantic model, and let every report use them. - Refresh in order, and check refresh history, because a silent failure shows old numbers. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is OneLake?

A A single report in Power BI

B The single logical data lake that all Fabric items read from

C A type of DAX measure

D A gateway for on-premises data

Answer: B. OneLake is the shared storage layer for Fabric.

2. When is a shortcut better than a copy?

A When you need an audit snapshot

B When you want the other location to read current data without duplicating storage

C When the source is offline

D Never; copies are always better

Answer: B. A shortcut points to the data, so it stays current.

3. Which item suits a team that writes T-SQL and needs stored procedures?

A A dataflow

B A warehouse

C A report

D A shortcut

Answer: B. Warehouses support full T-SQL, including stored procedures.

4. **Why define measures in the semantic model rather than in each report?**

A Reports cannot store measures

B So every report uses one definition, and a change is made once

C Measures run faster there only

D It hides the measures

Answer: B. One definition prevents numbers from drifting between reports.

5. **A dataflow fails overnight and nobody notices until a manager asks. What is the best fix?**

A Remove the dataflow

B Schedule refreshes in order and set a failure notification

C Refresh the report manually every hour

D Turn off the semantic model

Answer: B. Refresh order and failure alerts make silent failures visible.

EXERCISE

Design the chain

A company has order data in CSV files, customer and product lists in a CRM export, and three reports for finance, sales and operations. Draw the Fabric chain: which items you would use (lakehouse, warehouse, dataflow, notebook, semantic model, report), in what order, and where you would define the revenue and margin measures. Name one refresh rule and one governance control.

Show solution (worked answer)

One possible design: CSV files land in a lakehouse; a Dataflow Gen2 cleans and types them into tables; the tables are modelled as a star schema in a warehouse; one semantic model holds the relationships and the Total Sales and Gross Margin % measures; the three reports connect to that model. Refresh the dataflow first, then the model. Governance: endorse the model as Certified and limit edit rights to a named team.

POWER BI · LESSON 8

Databricks: Notebooks, Delta Lake, Spark Jobs and Unity Catalog

Work on a lakehouse with Databricks. Write notebooks in SQL and PySpark, store tables as Delta Lake with merges and time travel, run Spark jobs on a schedule, and govern data with Unity Catalog so that reports read trusted tables.

Gap-fill lesson | Databricks 2026 | Advanced | 150 min

By the end of this lesson you can

- Explain the Databricks workspace, clusters and notebooks
- Write a Delta table with SQL and PySpark, and explain why Delta adds reliability
- Use MERGE, time travel and schema checks on Delta tables
- Schedule a Spark job and understand how it differs from a local pandas script
- Organise data with Unity Catalog catalogs, schemas and permissions

What Databricks adds

Databricks is a data platform built on Apache Spark. Spark processes data in parallel across a cluster of machines, so it handles data too large for one computer. Databricks adds managed clusters, notebooks, scheduled jobs and a governance layer called Unity Catalog. Its storage format, **Delta Lake**, adds reliable updates and history to files stored in cloud object storage.

For a Power BI analyst the practical point is simple: many companies keep their reporting tables in Databricks, and reports connect to them through the SQL warehouse endpoint or through Power BI's Databricks connector.

CASE STUDY **Millions of rows, one daily refresh**

The sales file has grown to 80 million rows and no longer fits in a Power BI import. The data team keeps the detail in Delta tables in Databricks and builds a daily summary table with a scheduled Spark job. The report reads the summary, and the detail is queried only when someone drills down.

Workspace, clusters and notebooks

- **Workspace**: where notebooks, jobs and tables are organised.
- **Cluster** (or serverless compute): the machines that run your code. Clusters start, scale and stop; you pay while they run, so auto-termination matters.
- **Notebook**: a document of cells that mix Markdown, SQL, Python and PySpark. Each notebook attaches to a cluster and runs one cell at a time.

A notebook can switch language per cell with a magic command such as `%sql`, so one notebook can read a table in SQL and transform it in PySpark.

Reading data with PySpark

PySpark is the Python API for Spark. A DataFrame in Spark looks like a pandas DataFrame, but the work is split across the cluster and runs lazily: transformations are planned until a result is needed.

Python

```python
# In a Databricks notebook, `spark` is already created
sales = spark.read.table("retail.bronze.sales_raw")

summary = (
    sales
    .filter("order_date ≥ '2025-01-01'")
    .groupBy("region", "category")
    .agg({"sales": "sum", "profit": "sum"})
    .withColumnRenamed("sum(sales)", "total_sales")
    .withColumnRenamed("sum(profit)", "total_profit")
)

summary.write.mode("overwrite").saveAsTable("retail.gold.sales_summary_2025")
```

The names `retail.bronze.sales_raw` and `retail.gold.sales_summary_2025` follow the bronze, silver and gold layering: raw data, cleaned data and reporting data. This is the common pattern for lakehouse work.

::::note Lazy evaluation Calling `filter` or `groupBy` does not run anything. Spark runs the plan when you call an action such as `write`, `count` or `show`. A slow notebook is often a plan that recomputes data many times, which `cache()` or writing an intermediate table can fix. ::::

Delta Lake: reliable tables on files

A Delta table is a set of Parquet data files plus a transaction log that records every change. The log gives you:

- **ACID writes**: an update either completes or does not happen, so readers never see half-written data.
- **MERGE**: insert new rows and update changed ones in one statement, which is the upsert from the SQL incremental-loading lesson.
- **Time travel**: query the table as it was at an earlier version or time.
- **Schema enforcement**: writes with the wrong columns or types are rejected.

```sql
-- Upsert new and corrected order lines into the Delta table
MERGE INTO retail.silver.fact_sales AS target
USING retail.bronze.sales_today AS source
ON target.line_id = source.line_id
WHEN MATCHED THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *;

-- What did the table look like yesterday?
SELECT COUNT(*) AS lines, SUM(sales) AS total_sales
FROM retail.silver.fact_sales VERSION AS OF 12;

-- The history of changes
DESCRIBE HISTORY retail.silver.fact_sales;
```

Time travel is valuable for audits and for undoing a bad load: restore the previous version rather than rebuilding the table. Check retention settings first, because old versions are removed after a period.

::::warn MERGE needs a unique key If the source has two rows with the same `line_id`, MERGE fails or updates the same target row twice. Deduplicate the source before the merge, using the latest record for each key. ::::

Scheduled Spark jobs

A notebook becomes production work when it runs as a **job**:

1. Open the notebook, choose **Schedule** or create a job, and select a cluster or serverless compute.
2. Set the schedule (for example, every day at 5 a.m.) and the parameters.
3. Add a notification for failure, and set a retry for transient errors.
4. Check the run history after the first few runs, and compare the output totals with the source.

A job should be idempotent, like the loads in the SQL lesson: running it twice should not double the totals. Overwrite a summary table, or merge into a fact table, rather than appending.

Unity Catalog: governance across the platform

Unity Catalog is the central place to organise and secure data. Its three-level name is **catalog.schema.table**:

- **Catalog**: a top-level container, often one per environment or business unit (for example `retail`).
- **Schema**: a group of related tables (for example `gold` for reporting tables).
- **Table or view**: the data itself.

Permissions are granted on these objects, so a group can read `retail.gold` but not `retail.bronze`:

```sql
GRANT USE CATALOG ON CATALOG retail TO `analysts`;
GRANT USE SCHEMA ON SCHEMA retail.gold TO `analysts`;
GRANT SELECT ON SCHEMA retail.gold TO `analysts`;
```

Unity Catalog also records lineage, which shows which tables and notebooks produced a table. Use it to answer "where does this number come from?" before a report is trusted.

::::tip Give reports gold tables only Point Power BI at the gold layer, not at raw bronze tables. Gold tables are the cleaned, agreed definitions, so the report does not repeat cleaning logic and does not expose raw data to viewers. ::::

Connecting Databricks to Power BI

Power BI connects to a Databricks SQL warehouse through its Databricks connector, using the server hostname, HTTP path and an authentication method approved by the data team. Import mode suits daily summaries; DirectQuery suits large tables that must stay current. Choose the mode the same way as in the modes lesson: by data size, freshness and the number of users.

Comparing the platforms

Need	Fabric	Databricks
Storage	OneLake	Cloud storage with Delta tables
Large-scale code	Spark notebooks in a lakehouse	Spark notebooks and jobs
Governance	Workspaces, endorsement, lineage	Unity Catalog
Reporting	Native semantic models	Connect Power BI to a SQL warehouse

Both platforms run Spark, store tables in open formats, and feed Power BI. Learn the concepts once; the names change.

::::key What to remember - Databricks runs Spark on managed clusters; notebooks mix SQL and PySpark. - Delta Lake adds ACID writes, MERGE, time travel and schema checks to files. - Build summary tables as scheduled jobs, and make jobs safe to re-run. - Unity Catalog governs catalogs, schemas and tables with permissions and lineage. - Point reports at gold tables that hold agreed definitions. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is Delta Lake?

A A type of chart

B A table format that adds transaction logs, MERGE and time travel to files

C A Power BI gateway

D A programming language

Answer: B. Delta stores Parquet files with a log of every change.

2. What does time travel let you do?

A Make the cluster faster

B Query a table as it was at an earlier version

C Rename columns

D Skip a refresh

Answer: B. Time travel reads earlier table versions for audits and recovery.

3. Why must a MERGE source have unique keys?

A Keys are optional in Delta

B Duplicate keys in the source can make the merge fail or update one target row twice

C MERGE only works on text

D It speeds up the cluster

Answer: B. Deduplicate the source before merging.

4. In `retail.gold.sales_summary`, what is `gold`?

A The catalog

B The schema

C The table

D The cluster

Answer: B. The three-level name is catalog.schema.table.

5. Why should Power BI read gold tables rather than bronze?

A Bronze tables are faster to read

B Gold tables hold the cleaned, agreed definitions and avoid exposing raw data

C Gold tables cannot be queried

D Bronze tables are always deleted

Answer: B. Gold is the reporting layer.

EXERCISE **Design a daily job**

A retail team loads raw sales files each night into a bronze table. Design the daily job that produces a gold summary of sales by region and month for Power BI. Write the steps, state whether the summary is overwritten or merged and why, list one check to run before publishing, and name the Unity Catalog permission that lets the analysts read the summary but not the raw data.

Show solution (worked answer)

Steps: read the bronze table, deduplicate by line ID, filter the new period, aggregate sales and profit by region and month, write to the gold schema. Overwrite the summary because it is derived in full each run (re-runs give the same result). Check that the total sales in the summary equal the total in the cleaned silver table for the same period. Permission: `GRANT SELECT ON SCHEMA retail.gold TO analysts` with no grant on the bronze schema.